

Matter and antimatter in the dirac16complex theory: what can be proved

Exact charge conservation, discrete symmetries, the allowed charge-violating terms, the chirality pair of universes and the Sakharov conditions, with independent machine checks

dirac16complex project (with Claude Opus 5.5)

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1 Summary: the honest answer first

The honest answer. The request was to prove that this theory solves the current matter-antimatter mysteries; that statement is not proved in this document, and it cannot be proved, because within the theory as built its central part is false. The `dirac16complex` Lagrangian is exactly invariant under the phase transformation $\Psi \rightarrow e^{i\alpha}\Psi$ for every potential $U(S)$. This holds for anticommuting components (`dirac16complex`) and for commuting components (`dirac16complex00`) alike. The associated charge Q is therefore conserved in every gravitational field (for vanishing flux through the boundary of the slices), and no process described by the theory can create a net charge inside one universe; for the quantized field this holds at the formal level of Section 4.2. Sakharov's first condition fails for this charge. The theory also has exact symmetries of the Lagrangian that reverse the charge: C for the commuting field, in every gravitational field, and CP for the anticommuting field, in flat space and in gravitational fields with the corresponding reflection isometry. So Sakharov's second condition fails as well. No departure-from-equilibrium computation exists (the third condition). What can be proved is proved below exactly, with the machine checks named in each section (two independent implementations for M1 to M4 at the field and one-particle level). It is a precise account of matter and antimatter in the theory as it stands, together with a list of what would have to be added.

What is proved (theorems M1 to M4 at the field and one-particle level, the canonical structure of the image field in an exact Fock model, and the classical implication in M5; the Fock-level statements of M4 taken from Stage 5 are cited, Section 7.3):

- **M1 (exact U(1) and charge conservation).** For both statistics, every potential $U(S)$ and every gravitational field, the Lagrangian is invariant under $\Psi \rightarrow e^{i\alpha}\Psi$. The Noether current is $j^\mu = \bar{\Psi}\gamma^\mu\Psi$, the off-shell identity $\partial_\mu(\sqrt{|g|}j^\mu) = \sqrt{|g|}(\bar{E}\Psi + \bar{\Psi}E)$ holds, the current is conserved on shell, and the charge Q of every slice $x_4 = \text{const}$ is independent of x_4 (for vanishing boundary flux).
- **M2 (discrete symmetries).** All maps built from a constant 16×16 matrix, complex conjugation and reflections of frame directions are classified exactly, for both statistics, by one transformation rule. For the commuting field, C ($C_0 : \Psi \rightarrow \Psi^*$) is an exact symmetry in every gravitational field and reverses the charge. For the anticommuting field no constant C is a symmetry when $m \neq 0$, but $CP = C_8P$ ($C_8 : \Psi \rightarrow \gamma^8\Psi^*$, followed by the reflection of an odd number of the space-like directions $x_0, \dots, x_3$; C_0P is not exact) is an exact symmetry of the Lagrangian in flat space and in gravitational fields with the corresponding reflection isometry, and it reverses the charge. It preserves the canonical anticommutation relations as a linear (unitary-type) automorphism (derived); its implementation on the positive Fock space is not decided. In a generic gravitational field the background breaks it (Section 5.3).
- **M3 (charge-violating terms allowed by the symmetry).** The bilinears $\Psi^T M \Psi$ invariant under the identity component $\text{Spin}_0(4, 4)$ are exactly those with $M = C(\alpha P_- + \beta P_+)$, and all of them are symmetric. Hence the anticommuting field admits no Majorana-type mass term at all, while the commuting field admits two. The derivative bilinears are classified in the same way. Every such term carries $U(1)$ charge 2 and is absent from the Lagrangian.
- **M4 (the pair).** Under the chirality map $\Psi \rightarrow \gamma^8\Psi$ the current changes sign, and solutions with the parameters (m, λ) go to solutions with $(-m, -\lambda)$. A configuration Ψ_+ with (m, λ) and a second, independent classical field with the Lagrangian $\mathcal{L}_{-m, -\lambda}$ in the configuration $\Psi_- = \gamma^8\Psi_+$ have total charge 0 and total classical energy-momentum 0 in every gravitational

field. The one-particle (Krein) form is proved here. At the quantum level no reading gives a cancellation between two independent universes: the image field $\gamma^8\Psi_+$ is the same quantum system as Ψ_+ , with its own energy and charge equal to $+H_+$ and $+Q_+$ (proved here in an exact Fock model, Python only) and its own gravitational source equal to $+T_+$ (derived), and an independently quantised $-m$ field has positive energies. The Fock-level statements of Stage 5 are cited as a provisional Stage-5 result (Section 7.3).

- **M5 (a labelled hypothesis and its exact consequence).** If (H1) our universe is one member of a pair of two independent classical fields in the correlated configuration of M4 (the correlation is part of the assumption), (H2) the pair was created with $Q_+ = -Q_- \neq 0$, and (H3) the dirac16complex charge is identified with baryon number, then the total baryon number of the pair is zero and the excess of one member is balanced by the other. The implication is proved. H1, H2 and H3 are assumptions, none of them is derived, and the observed baryon-to-photon ratio $\eta \approx 6 \times 10^{-10}$ is not predicted. The balance is a property of the assumed configuration, not a compensation mechanism, and it has no quantum counterpart with two independent universes (Section 8.3).
- **M6 (Sakharov scorecard).** Condition 1 fails (M1), condition 2 fails for the Lagrangian (M2), condition 3 is not addressed (Section 9).

Labels used throughout. **Proved** means a complete argument is given here and the statement is also verified by named exact machine checks. **Derived** means that the statement follows from checked results by the argument given, but is not itself a machine check. **Recorded data** means floating-point numbers read from committed outputs. **Cited** means a result of another stage, taken from its reports, identified by their sha256 and not re-proved here. **Hypothesis** marks an assumption that nothing in this repository derives. Every number is copied from the reports listed in Section 11 or from the cited literature, except the one conversion in Section 2.2, which is marked as arithmetic done here.

2 The matter-antimatter problem from zero

2.1 Particles, antiparticles and conserved charges

Every known particle has an antiparticle with the same mass and the opposite charges: the electron has the positron, the proton the antiproton. A particle and its antiparticle can annihilate into radiation, and radiation of enough energy can create a particle-antiparticle pair. In both processes every conserved charge is unchanged, because the two members of a pair carry opposite charges.

Three charges matter here. The electric charge is exactly conserved. The baryon number B counts baryons (protons, neutrons and their relatives) as $+1$ and antibaryons as -1 ; each quark carries $1/3$. The lepton number L counts electrons, muons, tau leptons and neutrinos as $+1$ and their antiparticles as -1 . “Matter” in cosmology means baryons, and the matter-antimatter problem is the question why the universe contains baryons but essentially no antibaryons.

A charge is conserved when no process can change it. Noether’s theorem connects conservation to symmetry: if the action does not change under a continuous transformation that is the same at every point, there is a current j^μ with $\nabla_\mu j^\mu = 0$, and the integral of its time component over

space does not change with time. Theorem M1 is an instance of this theorem. It is the reason the dirac16complex theory, as built, cannot produce an asymmetry.

2.2 What is observed: the baryon asymmetry

The observed universe consists of matter. The size of the excess is expressed by the baryon-to-photon ratio

$$\eta = \frac{n_B - n_{\bar{B}}}{n_\gamma} \approx 6 \times 10^{-10},$$

the number density of baryons minus antibaryons divided by the number density of photons of the cosmic microwave background, roughly six baryons for every ten billion photons [5], [2]. It is measured in two independent ways. The abundances of the light elements (deuterium and helium in particular) produced in big-bang nucleosynthesis depend on η [5]. The acoustic peaks of the cosmic microwave background fix the baryon density; the Planck 2018 cosmological parameters give $\Omega_b h^2 = 0.0224 \pm 0.0001$ [2]. The two quantities are related through the photon number density of the microwave background at its present temperature T_0 ; Cyburt et al. [5] (their equation (11)) give

$$\eta_{10} \equiv 10^{10} \eta = 273.3036 \Omega_b h^2 (1 + 7.16958 \times 10^{-3} Y_p) \left(\frac{2.7255 \text{ K}}{T_0} \right)^3,$$

where Y_p is the primordial helium-4 mass fraction, whose small effect enters through the mean mass per baryon. With the Planck value and $Y_p \approx 0.25$ this gives $\eta \approx 6.1 \times 10^{-10}$ (arithmetic done here). The two determinations agree [5], [2].

Why is this a problem? If the hot early universe had contained exactly as many baryons as antibaryons, almost all of them would have annihilated, and far less matter than observed would remain. The excess could be put into the initial conditions of the universe, but the standard view, reviewed in [4], is that it should be generated by physical processes, from a state without asymmetry.

2.3 No antimatter domains

Could the universe be a patchwork of matter regions and antimatter regions, symmetric on the whole? Cohen, De Rújula and Glashow [6] showed that after recombination annihilation near the boundaries of such regions cannot be avoided, and that the gamma rays it produces would exceed the observed diffuse gamma-ray background unless the matter region we live in is essentially the entire visible universe. They concluded that a matter-antimatter symmetric universe of this kind is empirically excluded. Antimatter and matter also respond to gravity in the same direction as far as tested: antihydrogen atoms released in the ALPHA-g experiment move downward in the Earth's gravity, and repulsive "antigravity" is ruled out in that case [12]. This matters below, because the negative-mass member of the pair of Section 7 is not antimatter in our universe.

2.4 Sakharov’s three conditions

In 1967 Sakharov [1] identified three ingredients that any process generating the baryon asymmetry from a symmetric state must contain:

1. **Baryon-number violation.** Some process must change B ; otherwise $B = 0$ stays $B = 0$.
2. **C and CP violation.** C (charge conjugation) exchanges particles and antiparticles. If C were an exact symmetry, every process raising B would have a mirror process lowering B at the same rate, and the two would cancel. The same argument applies to CP, C combined with a spatial reflection P. Both C and CP must therefore be violated.
3. **Departure from thermal equilibrium.** In thermal equilibrium, with the baryon-number-violating reactions themselves in equilibrium, the chemical potential of B vanishes and the number of particles in each state is fixed by its energy alone. Particles and antiparticles have the same masses, so their equilibrium numbers are equal and the average B vanishes. An asymmetry can only be produced while the universe is out of equilibrium, for example during a phase transition or in the decay of heavy particles.

These three conditions are the yardstick of Section 9.

2.5 Why the Standard Model is not enough

The Standard Model contains all three ingredients in principle. Baryon number is violated at high temperature by electroweak sphaleron processes [7]. C is violated maximally by the weak interaction, and CP is violated by the complex phase of the quark mixing matrix. A departure from equilibrium could occur at the electroweak transition. The literature consensus, reviewed by Canetti, Drewes and Shaposhnikov [4], is nevertheless that the Standard Model cannot produce the observed asymmetry, for two reasons. First, lattice studies show that for Higgs masses of the order of the W-boson mass and above the electroweak transition is not a first-order phase transition, so it provides no strong departure from equilibrium [8]; the Higgs boson discovered at the LHC [10], [11] is heavier than the endpoint found in [8]. Second, the CP violation of the quark mixing matrix produces an asymmetry far too small [9], [13]. New physics beyond the Standard Model is therefore needed.

2.6 Pair-of-universes ideas

One class of ideas replaces “why is there more matter?” by a global symmetry: the universe as a whole is symmetric, but it consists of two parts, each of which is asymmetric. The known published example is the CPT-symmetric universe of Boyle, Finn and Turok [3]. They propose that the universe after the big bang is the CPT image of the universe before it, so that the pre-bang and post-bang epochs form a universe-antiuniverse pair, and they argue that CPT symmetry selects a vacuum state and gives a new interpretation of the cosmological baryon asymmetry. Nothing here depends on or tests that proposal. The pair of Section 7 is of a different kind: both members exist at the same time x_4 , and they are related by the chirality map with $m \rightarrow -m$, not by a time reflection through a bang. The resemblance is only structural: a global symmetry with local asymmetry arises here only under the hypotheses H1 and H2 of Section 8, and only for classical fields.

2.7 What a solution inside this theory would have to show

To solve the problem, a theory would have to show that a net baryon number can arise from a symmetric state, which requires the three conditions of Section 2.4, and it would have to compute η . Sections 4 to 9 show that the `dirac16complex` theory as built meets neither of the first two conditions: its charge is exactly conserved (M1), and it has exact charge-reversing symmetries (M2). What the theory does provide is an exact pairing of solutions, $(m, \lambda) \leftrightarrow (-m, -\lambda)$, with opposite charges (M4), which the pair-of-universes picture (H1) reads as two universes of masses $+M$ and $-M$. Here M is the universe mass of the author's notebook; the theorems use the mass parameter m of the Lagrangian, and " $\pm M$ universe" below means a solution with $\pm m$. This gives a conditional scenario (M5) that is stated here as a hypothesis, together with what would have to be added (Section 10).

3 The theory, its two fields and the conventions

The conventions are those of the Stage-1 document on `dirac16complex` in an arbitrary gravitational field and of the Stage-5 specification: indices count from 0; the coordinates are $x_0, \dots, x_7$, and x_4 is the evolution time; the tangent metric is $\eta = \text{diag}(+1, +1, +1, +1, -1, -1, -1, -1)$, so the directions 0, 1, 2, 3 are space-like and 4, 5, 6, 7 time-like. The gamma matrices $\gamma^0, \dots, \gamma^7$ are the notebook's split-octonion matrices, real signed permutation matrices with $\{\gamma^a, \gamma^b\} = 2\eta^{ab}$, stored in the exact fixture artifacts/`dirac16complex`/arbitrary-field/algebra-fixture.json. Further

$$\begin{aligned} C &= \gamma^0 \gamma^1 \gamma^2 \gamma^3 = C^T = C^{-1}, & \bar{\Psi} &= \Psi^\dagger C, & (C\gamma^a)^T &= -C\gamma^a, \\ \gamma^8 &= \gamma^0 \gamma^1 \dots \gamma^7 = \text{diag}(-I_8, I_8), & P_\mp &= \tfrac{1}{2}(1 \mp \gamma^8), & B &= -iC\gamma^4, \\ S^{ab} &= \tfrac{1}{4}[\gamma^a, \gamma^b], & \Omega_\mu &= \tfrac{1}{2}\omega_{\mu ab}S^{ab}, & D_\mu \Psi &= \partial_\mu \Psi + \Omega_\mu \Psi, & D_\mu \bar{\Psi} &= \partial_\mu \bar{\Psi} - \bar{\Psi} \Omega_\mu. \end{aligned}$$

Here $\omega_{\mu ab} = \eta_{ac} \omega_\mu{}^c{}_b$ is the canonical (Levi-Civita) spin connection of the vielbein $e_\mu{}^a$, and $\gamma^\mu = e_a{}^\mu \gamma^a$.

The two fields. Both are 16-component complex fields $\Psi = (\Psi_0, \dots, \Psi_{15})^T$ carrying the irreducible complex Clifford module of $\text{Pin}(4,4)$. For **dirac16complex** the components are anticommuting (Grassmann-odd): $\Psi_a \Psi_b = -\Psi_b \Psi_a$, $\Psi_a \Psi_b^* = -\Psi_b^* \Psi_a$, and complex conjugation reverses products, $(\theta_1 \theta_2)^* = \theta_2^* \theta_1^*$. For **dirac16complex00** the components are commuting complex numbers. We write $s = -1$ for anticommuting and $s = +1$ for commuting components (the statistics sign).

The Lagrangian (L1 of the Stage-5 specification), for both fields:

$$\begin{aligned} \mathcal{L} &= \sqrt{|g|} [K - mS - U(S)], & S &= \bar{\Psi} \Psi, \\ K &= \tfrac{1}{2}(\bar{\Psi} \gamma^\mu D_\mu \Psi - (D_\mu \bar{\Psi}) \gamma^\mu \Psi), & U(S) &= \tfrac{\lambda}{2} S^2 \text{ (default)}. \end{aligned}$$

For anticommuting components U must be a polynomial in S of degree at most 16, because $S^{17} = 0$; for commuting components U may be any function. $\mathcal{L}_{m,\lambda}$ denotes the Lagrangian with mass m and $U = \frac{\lambda}{2} S^2$.

The field equations are $E = 0$ and $\bar{E} = 0$ with

$$E := \gamma^\mu D_\mu \Psi - (m + U'(S)) \Psi, \quad \bar{E} := (D_\mu \bar{\Psi}) \gamma^\mu + (m + U'(S)) \bar{\Psi},$$

derived in Stage 1 (Section 8) and re-verified there in a genuine Grassmann algebra.

Current and charge. The current is $j^\mu = \bar{\Psi}\gamma^\mu\Psi$. Because $C\gamma^a$ is real and antisymmetric, j^μ is anti-Hermitian, and the Hermitian current of Stage 1 is $J^\mu = -ij^\mu$. The charge of the slice $\Sigma_{x_4} = \{x_4 = \text{const}\}$ is

$$Q(x_4) = \int_{\Sigma_{x_4}} \sqrt{|g|} J^{x_4} d^7x.$$

In Gaussian normal gauge ($g^{44} = -1$, $\gamma^{x_4} = \gamma^4$) the charge density is $J^4 = \Psi^\dagger B\Psi$. The matrix B is Hermitian with $B^2 = 1$ and has 8 eigenvalues $+1$ and 8 eigenvalues -1 , so Q is an indefinite quadratic form. After quantization B is the Krein metric of the canonical anticommutator $\{\Psi, \Psi^\dagger\} = B\delta^7/\sqrt{|g|}$, and in the good sector particles carry $Q = +1$ and antiparticles $Q = -1$ (Stage 1, Sections 10.5 to 10.9).

Test geometries. The Wolfram checks use G1, the generic non-diagonal vielbein of Stage 1 at its three rational points p1, p2, p3. The Python checks use two fields built independently from vielbein jets at one point: G_A, a non-diagonal vielbein with space-space, space-time and time-time mixing, and G_B, a diagonal vielbein depending on x_0 and x_4 only, for which every reflection of $x_1, x_2, x_3, x_5, x_6, x_7$ is an isometry. The Python checker also proves the one geometric input of M1 for arbitrary first vielbein jets (Section 4.2).

4 Theorem M1: exact U(1) symmetry and charge conservation

4.1 Statement

Theorem M1. For both statistics, every potential U (a polynomial of degree at most 16 in S for anticommuting components, an arbitrary function for commuting ones) and every vielbein:

1. (global phase) $\mathcal{L}[e^{i\alpha}\Psi] = \mathcal{L}[\Psi]$ for every constant real α ;
2. (local phase and Noether current) for a phase $\alpha(x)$ that depends on the point, $\mathcal{L}[e^{i\alpha}\Psi] = \mathcal{L}[\Psi] + i\sqrt{|g|}(\partial_\mu\alpha)j^\mu$ with $j^\mu = \bar{\Psi}\gamma^\mu\Psi$, so the Noether current is $N^\mu = i\sqrt{|g|}j^\mu$; it equals the Noether formula $N^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\Psi_a)}(i\Psi_a) + (-i\Psi_a^\dagger)\frac{\partial\mathcal{L}}{\partial(\partial_\mu\Psi_a^\dagger)}$ (right derivative for Ψ and left derivative for $\Psi^\dagger$ in the anticommuting case);
3. (off-shell identity) with E and $\bar{E}$ of Section 3, $\partial_\mu(\sqrt{|g|}j^\mu) = \sqrt{|g|}(\bar{E}\Psi + \bar{\Psi}E)$;
4. (conservation) on every solution $\nabla_\mu j^\mu = |g|^{-1/2}\partial_\mu(\sqrt{|g|}j^\mu) = 0$, and $Q(x_4)$ does not depend on x_4 whenever the flux of $\sqrt{|g|}j^\mu$ through the boundary of the slices vanishes (compact support, sufficient fall-off, or periodic identification of the slice coordinates);
5. (charge density) in Gaussian normal gauge $J^4 = \Psi^\dagger B\Psi$, with B Hermitian, $B^2 = 1$, spectrum $(+1)^8(-1)^8$ and trace 0.

Consequence. No solution of the field equations of either field, for any U and in any gravitational field with $g^{44} \neq 0$, changes Q (for vanishing boundary flux, as in item 4). The charge of a universe is fixed by its initial data at $x_4 = 0$.

4.2 Proof

Item 1. For commuting components the phase multiplies Ψ by $e^{i\alpha}$ and $\Psi^\dagger$ by $e^{-i\alpha}$. For anticommuting components complex conjugation is antilinear, $(c\theta)^* = \bar{c}\theta^*$, so the substitution $\Psi \rightarrow e^{i\alpha}\Psi$, $\Psi^* \rightarrow e^{-i\alpha}\Psi^*$ is an automorphism of the Grassmann algebra that preserves the order of every product. Every term of $\mathcal{L}$ is a sum of products that contain as many factors from $\Psi^\dagger$ (or $\partial\Psi^\dagger$) as factors from Ψ (or $\partial\Psi$): K and S are bilinears $\Psi^\dagger X\Psi$ whose geometric matrices X are untouched by the phase, and $U(S)$ is a function of S . For anticommuting components S^k has exactly $\binom{16}{k}$ monomials, each with k factors Ψ^* and k factors Ψ , and $S^{17} = 0$, so every admissible U is a polynomial in S of degree at most 16 and has charge 0. Hence every monomial of $\mathcal{L}$ is multiplied by $e^{i\alpha}e^{-i\alpha} = 1$.

Item 2. With $\alpha = \alpha(x)$ one has $D_\mu(e^{i\alpha}\Psi) = e^{i\alpha}(D_\mu\Psi + i(\partial_\mu\alpha)\Psi)$ and $D_\mu(\bar{\Psi}e^{-i\alpha}) = e^{-i\alpha}(D_\mu\bar{\Psi} - i(\partial_\mu\alpha)\bar{\Psi})$, while S and $U(S)$ are unchanged. Therefore

$$K \rightarrow K + \frac{1}{2}(i(\partial_\mu\alpha)\bar{\Psi}\gamma^\mu\Psi + i(\partial_\mu\alpha)\bar{\Psi}\gamma^\mu\Psi) = K + i(\partial_\mu\alpha)j^\mu,$$

and $\mathcal{L}$ changes by $i\sqrt{|g|}(\partial_\mu\alpha)j^\mu$. The coefficient of $\partial_\mu\alpha$ is the Noether current. The Noether formula gives the same expression, because $\partial\mathcal{L}/\partial(\partial_\mu\Psi) = \frac{1}{2}\sqrt{|g|}\bar{\Psi}\gamma^\mu$ and $\partial\mathcal{L}/\partial(\partial_\mu\Psi^\dagger) = -\frac{1}{2}\sqrt{|g|}C\gamma^\mu\Psi$ with the derivative conventions of item 2.

Item 3. The Leibniz rule gives

$$\begin{aligned} \partial_\mu(\sqrt{|g|}\bar{\Psi}\gamma^\mu\Psi) &= \sqrt{|g|}(\partial_\mu\bar{\Psi})\gamma^\mu\Psi + \bar{\Psi}\partial_\mu(\sqrt{|g|}\gamma^\mu\Psi) \\ &\quad + \sqrt{|g|}\bar{\Psi}\gamma^\mu\partial_\mu\Psi. \end{aligned}$$

The only geometric input is the divergence identity of Stage 1 (Result 5.3), a consequence of the covariant constancy of the gammas:

$$\partial_\mu(\sqrt{|g|}\gamma^\mu) = \sqrt{|g|}[\gamma^\mu, \Omega_\mu] = \sqrt{|g|}(\gamma^\mu\Omega_\mu - \Omega_\mu\gamma^\mu).$$

Substituting it and collecting terms,

$$\begin{aligned} \partial_\mu(\sqrt{|g|}j^\mu) &= \sqrt{|g|}[(\partial_\mu\bar{\Psi} - \bar{\Psi}\Omega_\mu)\gamma^\mu\Psi + \bar{\Psi}\gamma^\mu(\partial_\mu\Psi + \Omega_\mu\Psi)] \\ &= \sqrt{|g|}[(D_\mu\bar{\Psi})\gamma^\mu\Psi + \bar{\Psi}\gamma^\mu D_\mu\Psi]. \end{aligned}$$

On the other hand

$$\bar{E}\Psi + \bar{\Psi}E = (D_\mu\bar{\Psi})\gamma^\mu\Psi + (m + U'(S))S + \bar{\Psi}\gamma^\mu D_\mu\Psi - (m + U'(S))S,$$

because $U'(S)$ is even and commutes with every component. The two potential terms cancel, and item 3 follows. No factor was reordered, so the argument holds verbatim for both statistics.

Item 4. On shell $E = \bar{E} = 0$, so $\partial_\mu(\sqrt{|g|}j^\mu) = 0$, and for a vector field $\nabla_\mu j^\mu = |g|^{-1/2}\partial_\mu(\sqrt{|g|}j^\mu)$. Integrate $\partial_\mu(\sqrt{|g|}J^\mu) = 0$ over the slab between the slices $x_4 = t_1$ and $x_4 = t_2$. The divergence theorem in the coordinates $x_0, \dots, x_7$ (pure calculus, valid for any signature) gives

$$Q(t_2) - Q(t_1) = - \int_{t_1}^{t_2} dx_4 \oint \sqrt{|g|} J^i dS_i,$$

the flux through the lateral boundary, which vanishes under the stated conditions. The indefiniteness of J^4 plays no role: Q is conserved although it is not positive.

Item 5. $\sqrt{|g|} J^{x_4} = -i\sqrt{|g|} \Psi^\dagger C \gamma^{x_4} \Psi$, and in Gaussian normal gauge $-iC\gamma^4 = B$. The properties of B are those of Stage 1 (Result 10.2); they are re-verified here, including the Hermiticity of the charge matrix $-i\sqrt{|g|} C \gamma^{x_4}$ in a general frame.

The quantized field. The identity of item 3 is algebraic, and in it every $\Psi^\dagger$ stands to the left of every Ψ , so it holds for the operator-valued field as well; the Heisenberg equations of the quantized theory are the field equations (Stage 1, Section 10.4), and normal ordering changes Q by a constant. Hence the normal-ordered charge is conserved in the quantized theory too (derived and formal, not a separate machine check: no regularization is constructed). A classically conserved current can fail to be conserved after quantization through an anomaly, and this is how baryon number is violated in the Standard Model (Section 2.5, [7]). No anomaly of that kind is expected here: the $U(1)$ acts in the same way on both chiralities (it is vector-like), no gauge field couples to j , and the $U(1)$ -invariant mass term $m\bar{\Psi}\Psi$ would permit a $U(1)$ -preserving regularization of Pauli-Villars type. This is an argument, not a proof; a regularized quantum theory in signature (4,4) is not constructed here (non-claim 8). $\square$

4.3 How M1 was verified

Wolfram (wolfram/Dirac16ComplexMatterAntimatter.wl; anticommuting components in the package's exact Grassmann algebra with 1440 odd generators for the values and the first and second derivatives of Ψ and $\Psi^\dagger$; commuting components as exact symbols):

- Commuting components with an undefined function U , in flat space and at the three G1 points: the argument of U is identically invariant, so $U(S)$ is invariant for every function U .
- Anticommuting components at the three G1 points with $U = \frac{\lambda}{2} S^2 + \frac{c_3}{3} S^3$: the Lagrangian density has 1848 monomials at each point, all of $U(1)$ charge 0, and the explicit phase automorphism leaves it unchanged. The powers S^k , $k = 1, \dots, 17$, have 16, 120, 560, 1820, 4368, 8008, 11440, 12870, 11440, 8008, 4368, 1820, 560, 120, 16, 1 and 0 monomials, with coefficients $\pm k!$ and charge 0.
- The local-phase variation and the Noether formula both give $i\sqrt{|g|} \bar{\Psi} \gamma^\mu \Psi$ at the three G1 points, for both statistics.
- The off-shell identity of item 3 holds exactly at the three G1 points: for anticommuting components with 1088 monomials on the left-hand side and the potential terms cancelling; for commuting components with an undefined function U' .
- Negative control: with the notebook's contraction of the spin connection (Stage 1, Section 5.6) the identity fails at all three points, as it must, since the divergence identity fails for that contraction.

- On shell: with exact on-shell jets from the Stage-1 solver at the three G1 points, for $(m, \lambda) = (3/7, 5/11)$, $(-2/5, 7/13)$ and $(5/9, -3/8)$, the field equations vanish at the point and so does the divergence of the current, while for generic off-shell data the divergence does not vanish.

Python (scripts/check_dirac16complex_matter_antimatter.py, independent; anticommuting components with scripts/grassmann_algebra.py, commuting components with its own exact polynomial class; exact phase $e^{i\alpha} = (3 + 4i)/5$; $m = 3/2$, $\lambda = 5/7$ and $U = \frac{\lambda}{2}S^2 + \frac{\mu_3}{3}S^3$ with $\mu_3 = 2/3$):

- In G_A and in G_B, for both statistics: finite U(1) invariance, the Noether current $N^\mu = i\sqrt{|g|} \bar{\Psi}\gamma^\mu\Psi$, the Euler-Lagrange expressions computed directly and in closed form, the off-shell Noether identity, the current divergence identity and Hermiticity, all as exact polynomial identities in fully generic field jets. The Lagrangian density has 1364 (anticommuting) and 1636 (commuting) monomials in G_A and 952 and 1224 in G_B; the Noether divergence has 592 monomials in G_A and 272 in G_B.
- The divergence identity $\partial_\mu(\sqrt{|g|}\gamma^\mu) = \sqrt{|g|}[\gamma^\mu, \Omega_\mu]$ and $D_\mu\gamma^\nu = 0$ are proved for arbitrary first vielbein jets. At a point both sides are linear in the 512 numbers $\partial_\lambda e_\nu^a$, and all 512 basis directions are verified exactly (448 of them have $\Omega \neq 0$), for the identity frame and for a unimodular non-diagonal frame; a constant change of coordinates brings any frame at a point to such a frame. With the notebook contraction both identities fail (negative control).

checks (Wolfram)

```
MA_M1_u1InvarianceCommutingGenericU_flat  MA_M1_u1InvarianceCommutingGenericU_G1
MA_M1_u1InvarianceGrassmann_G1  MA_M1_grassmannPotentialsPolynomialAndNeutral
MA_M1_noetherCurrentLocalPhase_commuting_G1
MA_M1_noetherCurrentLocalPhase_grassmann_G1
MA_M1_noetherCurrentFormula_grassmann_G1  MA_M1_noetherCurrentFormula_commuting_G1
MA_M1_noetherIdentity_grassmann_G1  MA_M1_noetherIdentity_commuting_G1
MA_M1_negativeControlNotebookConnection  MA_M1_onShellConservation_G1
MA_M1_chargeDensityMatrix  MA_M1_stage1ChecksCited
checks (Python), for G in {G_A_generic_nondiagonal, G_B_diagonal_x0_x4} and
X in {grassmann, commuting}: MA_M1_<name>_<X>_<G> with <name> one of
    u1Invariance  noetherCurrentIsISqrtgPsiGammaPsi  eulerLagrangeCrossCheck
    eulerLagrangeClosedForm  noetherIdentity  currentDivergenceIdentity  hermiticity
and MA_M1_geometry_<G>  MA_M1_divergenceIdentityAllFirstJets  MA_M1
```

The check MA_M1_stage1ChecksCited cites, by check name and report hash, the Stage-1 results used here: the Euler-Lagrange equations for both variations with $\lambda \neq 0$ in a genuine Grassmann algebra, the divergence identity, local spin invariance, the conservation of the energy-momentum tensor, the Hermiticity of the current, the chirality map, the Pin characters, the invariant forms and the charge form B . The cited Stage-1 checks are

```
wolfram-geometry-report.json  GEO_divergenceIdentity_G1  GEO_divergenceIdentity_G2
LAG_eulerLagrangePsiBar_G1  LAG_eulerLagrangePsiBar_G2  LAG_eulerLagrangePsi_G1
LAG_eulerLagrangePsi_G2  LAG_localSpinInvariance_G1  LAG_localSpinInvariance_G2
EMT_conservation_G1  EMT_conservation_G2
wolfram-algebra-report.json  QNT_currentHermiticity  ALG_gamma8Map
ALG_pinLiftCharacter  ALG_invariantForms  ALG_chargeFormB
grassmann-demo-report.json  GR_currentHermitian
```

4.4 The Kohn-Sham states have a fixed particle number (recorded data)

The Kohn-Sham (density-functional) states of Stages 4 and 5 are computed at a fixed particle number N : the Mermin functional contains the constraint $\sum_n f_n = N$ with the chemical potential μ as its Lagrange multiplier, and in the no-sea convention of Stage 4 the net occupation (particle occupations minus sea holes) is the Kohn-Sham image of the $U(1)$ charge. These states are therefore, by construction, states of one fixed charge. As a data check, not a proof, the Wolfram verifier reads the committed Stage-4 runs (check `MA_M1_ksFixedNetNumberRecorded`): 56 reference-solver run files with 168 levels and 33 Rust run files all have net occupation N to 10^{-9} relative, with largest absolute deviations $3.5073\text{e-}10$ (reference solver) and $1.012\text{e-}11$ (Rust). This is recorded floating-point data, and it is the only check quoted in this document that is not exact.

5 Theorem M2: charge conjugation, reflections and time reversal

5.1 The maps

For a subset $R \subseteq \{0, \dots, 7\}$ of directions let $r_a = -1$ for $a \in R$ and $r_a = +1$ otherwise, and let $x \mapsto Rx$ be the reflection $x_a \mapsto r_a x_a$. For a constant invertible 16×16 matrix M consider the linear and the antilinear maps

$$T_{M,R}^{\text{lin}} : \Psi'(x) = M \Psi(Rx), \quad T_{M,R}^{\text{anti}} : \Psi'(x) = M \Psi^*(Rx).$$

For $R = \emptyset$ the antilinear maps are the charge conjugations; for $R \neq \emptyset$ and M a $\text{Pin}(4,4)$ element they are reflections (P-type if $R \subseteq \{0, 1, 2, 3\}$, T-type if $4 \in R$) and their combinations with C. The form $\Psi' = M \bar{\Psi}^T$ of the specification is included: since $C^T = C$, $\bar{\Psi}^T = C \Psi^*$, so $M \bar{\Psi}^T = (MC) \Psi^*$. We write $s_R = |R \cap \{0, 1, 2, 3\}|$ for the number of reflected space-like directions, $\sigma_R = (-1)^{s_R}$, and $\Gamma_A = \gamma^{a_1} \gamma^{a_2} \dots \gamma^{a_k}$ ($a_1 < \dots < a_k$) for the Clifford monomial of a set A , with $\Gamma_\emptyset = 1$ and R^c the complement of R .

5.2 Three lemmas

Lemma A (charge-conjugation intertwiners). The solutions of $\gamma^a M = \eta M (\gamma^a)^*$ for all a are $M \in \mathbb{C} \cdot 1$ for $\eta = +1$ and $M \in \mathbb{C} \cdot \gamma^8$ for $\eta = -1$. The solutions of $\gamma^a M = \zeta M (\gamma^a)^T$ are $M \in \mathbb{C} \cdot C$ for $\zeta = -1$ and $M \in \mathbb{C} \cdot \gamma^8 C$ for $\zeta = +1$. Hence every constant charge conjugation compatible with the Dirac operator is, up to a factor, $C_0 : \Psi \rightarrow \Psi^*$ or $C_8 : \Psi \rightarrow \gamma^8 \Psi^*$, and the forms $C \bar{\Psi}^T = \Psi^*$ and $\gamma^8 C \bar{\Psi}^T = \gamma^8 \Psi^*$ are the same two maps.

Proof. The gammas are real, so the first condition reads $\gamma^a M = \eta M \gamma^a$. For $\eta = +1$, M commutes with every γ^a , and the commutant of the gammas consists of the scalars (Stage 1, Result 4.1: $\mathbb{C}^{16}$ is an irreducible $\text{Pin}(4,4)$ module). For $\eta = -1$, γ^8 anticommutes with every γ^a , so $\gamma^8 M$ commutes with every γ^a and is a scalar. For the transposed condition, $(C \gamma^a)^T = -C \gamma^a$ gives $(\gamma^a)^T = -C \gamma^a C^{-1}$, so $\gamma^a M = \zeta M (\gamma^a)^T$ becomes $\gamma^a (MC) = -\zeta (MC) \gamma^a$, and the first part applies to MC . $\square$

Lemma B (sign patterns). For every R the equation $M^{-1} \gamma^a M r_a = \varepsilon \gamma^a$ (all a , one common sign ε) has, up to a factor, exactly two solutions: $M = \Gamma_R$ with $\varepsilon = (-1)^{|R|}$ and $M = \Gamma_{R^c}$ with

$\varepsilon = -(-1)^{|R|}$. Both satisfy $M^\dagger = M^T = M^{-1}$ and

$$M^\dagger C M = \sigma_R C, \quad \sigma_R = (-1)^{s_R}.$$

Proof. γ^a anticommutes with γ^b for $b \neq a$ and commutes with itself, so $\Gamma_A^{-1} \gamma^a \Gamma_A = (-1)^{|A|} \gamma^a$ for $a \notin A$ and $(-1)^{|A|-1} \gamma^a$ for $a \in A$. For $A = R$ this is $(-1)^{|R|} r_a \gamma^a$, for $A = R^c$ it is $-(-1)^{|R|} r_a \gamma^a$. The 256 monomials realise the 256 sign patterns $\gamma^a \mapsto \pm \gamma^a$ bijectively, and two solutions of the same pattern differ by an element of the commutant, a scalar (Lemma A). The gammas are real signed permutation matrices, hence orthogonal, and so are their products: $M^\dagger = M^T = M^{-1}$. Finally $(\gamma^a)^T C \gamma^a = -C$ for $a \leq 3$ (γ^a symmetric, anticommuting with C , square +1) and $(\gamma^a)^T C \gamma^a = +C$ for $a \geq 4$ (γ^a antisymmetric, commuting with C , square -1). Applying this factor by factor, $\Gamma_A^T C \Gamma_A = (-1)^{|A \cap \{0,1,2,3\}|} C$; for $A = R$ and $A = R^c$ the exponents are s_R and $4 - s_R$, of the same parity. $\square$

Lemma C (statistics sign). For every matrix Y , $\Psi^T Y \Psi^* = s \Psi^\dagger Y^T \Psi$, with $s = -1$ for anticommuting and $s = +1$ for commuting components; the same holds with derivatives on either factor.

Proof. $\Psi^T Y \Psi^* = \sum_{ij} \Psi_i Y_{ij} \Psi_j^* = s \sum_{ij} \Psi_j^* Y_{ij} \Psi_i = s \Psi^\dagger Y^T \Psi$, because exchanging two odd factors costs a sign and exchanging two commuting factors does not. $\square$

5.3 The transformation rule

Theorem M2 (transformation rule). Let M be one of the two solutions of Lemma B for R , with its sign ε (a factor of modulus 1 is irrelevant). Then, for $U = \frac{\lambda}{2} S^2$,

$$\mathcal{L}_{m,\lambda}[T\Psi](x) = \kappa \mathcal{L}_{\sigma\kappa m, \kappa\lambda}[\Psi](Rx), \quad j'^a(x) = c_j r_a j^a(Rx),$$

with

$$\begin{aligned} \text{linear maps:} \quad & \kappa = \sigma_R \varepsilon, & \sigma = \sigma_R, & c_j = \kappa, \\ \text{antilinear maps:} \quad & \kappa = s \sigma_R \varepsilon, & \sigma = s \sigma_R, & c_j = -\kappa. \end{aligned}$$

Here κ is the sign of the kinetic term ($K \rightarrow \kappa K$), σ the sign of the scalar ($S \rightarrow \sigma S$), and c_j the sign of the current relative to the vector law $j^a \rightarrow r_a j^a$.

Proof, linear maps (flat space). $\partial_a \Psi'(x) = r_a M(\partial_a \Psi)(Rx)$ and $\Psi'^\dagger = \Psi^\dagger M^\dagger$. By Lemma B, $M^\dagger C \gamma^a M = (M^{-1} C M)(M^{-1} \gamma^a M) = \sigma_R \varepsilon r_a C \gamma^a$. Hence $S \rightarrow \sigma_R S$, $j^a \rightarrow \sigma_R \varepsilon r_a j^a$, and in K every term acquires $\sigma_R \varepsilon r_a \cdot r_a = \sigma_R \varepsilon$. Then

$$\kappa K - m \sigma S - \frac{\lambda}{2} S^2 = \kappa [K - (\sigma \kappa m) S - \kappa \frac{\lambda}{2} S^2] = \kappa \mathcal{L}_{\sigma\kappa m, \kappa\lambda} / \sqrt{|g|},$$

using $\kappa^2 = 1$ and $(\sigma S)^2 = S^2$.

Proof, antilinear maps (flat space). Now $\Psi'^\dagger = \Psi^T M^\dagger$ (conjugation is antilinear and $(M \Psi^*)^* = M^* \Psi$). By Lemma C and Lemma B,

$$S' = \Psi^T (M^\dagger C M) \Psi^* = s \Psi^\dagger (\sigma_R C)^T \Psi = s \sigma_R S,$$

and with $Y = M^\dagger C \gamma^a M = \sigma_R \varepsilon r_a C \gamma^a$ and $(C \gamma^a)^T = -C \gamma^a$,

$$\begin{aligned} \Psi'^\dagger C \gamma^a \partial_a \Psi' &= r_a \Psi^T Y \partial_a \Psi^* = r_a s (\partial_a \Psi^\dagger) Y^T \Psi = -s \sigma_R \varepsilon (\partial_a \Psi^\dagger) C \gamma^a \Psi, \\ -(\partial_a \Psi'^\dagger) C \gamma^a \Psi' &= -r_a (\partial_a \Psi^T) Y \Psi^* = -r_a s \Psi^\dagger Y^T \partial_a \Psi = s \sigma_R \varepsilon \Psi^\dagger C \gamma^a \partial_a \Psi. \end{aligned}$$

So $K \rightarrow s \sigma_R \varepsilon K$, and in the same way $j^a \rightarrow -s \sigma_R \varepsilon r_a j^a$. The Lagrangian follows as in the linear case.

Curved fields. In a curved field the same maps are used in frame form: $\Psi \rightarrow M \Psi$ (or $M \Psi^*$) together with the constant frame change $e_\mu^a \rightarrow r_a e_\mu^a$ (no sum over a), which leaves the metric unchanged. Then $\gamma'^\mu = \sum_a r_a e_a^\mu \gamma^a$ and $\omega'_{\mu ab} = r_a r_b \omega_{\mu ab}$. Lemma B gives $M[\gamma^a, \gamma^b] M^{-1} = r_a r_b [\gamma^a, \gamma^b]$, hence $\Omega'_\mu = M \Omega_\mu M^{-1}$ and $D'_\mu (M \Psi) = M D_\mu \Psi$. For antilinear maps Ω_μ is real, so $(D_\mu \Psi)^* = D_\mu \Psi^*$. With these replacements the flat computation goes through unchanged, with ∂_a replaced by D_μ , and the rule holds in every gravitational field. When the metric has a reflection isometry, the coordinate form $\Psi(Rx)$ holds as well. $\square$

Scope in a gravitational field. The frame form changes the background frame, $e \rightarrow e \text{diag}(r)$. When $R = \emptyset$ (the internal maps C_0 , C_8 and the chirality map) nothing changes, so C_0 is an exact symmetry of the commuting field in every gravitational field (check MA_M2_internalMapsCurvedJets). The exact CP maps of the anticommuting field have s_R odd (Corollary M2.1 below). Then $\text{diag}(r)$ has space-block determinant -1 and lies outside $\text{SO}_0(4, 4)$: it is not a local Lorentz transformation, and no linear Pin lift of it is an exact symmetry that could undo it. For these maps the rule relates the theory on the frame e to the theory on the frame $e \text{diag}(r)$. It is a symmetry of a fixed background only when the metric has the corresponding reflection isometry, as in flat space, in the primordial field for reflections of x_1, x_2, x_3 , and in G_B. At a fixed frame of the generic field G_A, which has no reflection isometry, the internal part $\Gamma_{R^c} \Psi^*$ of $C_8 P_b$ ($b = 0, \dots, 3$) and of $C_8 P_{123}$ maps $\mathcal{L}$ to none of $\pm \mathcal{L}_{\pm m, \pm \lambda}$ (check MA_M2_cpScopeInCurvedFields, which also verifies the determinant statement for all 128 sets R with s_R odd and that no linear Pin lift of such a reflection is exact). A generic gravitational field therefore breaks CP of the anticommuting field. This does not open a way to an asymmetry, because the charge is conserved in every gravitational field (M1).

5.4 Classification

Corollary M2.1 (exact symmetries). A map of Section 5.1 is an exact symmetry, $\mathcal{L}[T\Psi](x) = \mathcal{L}[\Psi](Rx)$ for all m and λ , if and only if $\kappa = \sigma = 1$. Then it is a symmetry for every potential U , because $S \rightarrow S$ and $K \rightarrow K$. Explicitly:

- linear maps: an exact symmetry with reflected set R exists if and only if s_R is even, for any number of reflected time-like directions; it is the monomial among Γ_R, Γ_{R^c} with $\varepsilon = +1$;
- antilinear maps, commuting components: if and only if s_R is even;
- antilinear maps, anticommuting components: if and only if s_R is odd.

No map of Section 5.1 sends $\mathcal{L}_{m,\lambda}$ to $-\mathcal{L}_{m,\lambda}$ when $\lambda \neq 0$: that would need $\kappa = -1$ together with $\sigma\kappa = 1$ and $\kappa\lambda = \lambda$. For $\lambda = 0$ ($U = 0$) the maps of the class $-\mathcal{L}_{m,-\lambda}$ do so, and they are then symmetries of the field equations, which are the same for $\mathcal{L}$ and $-\mathcal{L}$ (derived from the rule).

Corollary M2.2 (charge). For an exact symmetry, $c_j = +1$ for linear and $c_j = -1$ for antilinear maps. If x_4 is not reflected ($r_4 = +1$), an exact linear symmetry preserves j^4 and Q , and an exact antilinear symmetry reverses them, $Q \rightarrow -Q$.

Counting. Per statistics there are 256 sets R , 2 matrices and 2 types, 1024 maps. Their Lagrangian images fall into the four classes $\mathcal{L}_{m,\lambda}$, $\mathcal{L}_{-m,\lambda}$, $-\mathcal{L}_{m,-\lambda}$ and $-\mathcal{L}_{-m,-\lambda}$ with 256 maps each. The exact symmetries number 256 for each statistics: 128 linear (the sets with s_R even) and 128 antilinear. Among them, the charge-reversing ones that do not reflect x_4 are the exact antilinear maps with $4 \notin R$. There are $8 \times 8 = 64$ of them for each statistics: $R \cap \{0, 1, 2, 3\}$ of even size (commuting) or odd size (anticommuting), times any subset of $\{5, 6, 7\}$.

5.5 C, P, CP, T and CPT for the two fields

The table applies the rule to the named maps. $R_A x$ denotes x with the coordinates x_a , $a \in A$, reversed, and Γ_A is the Clifford monomial of Section 5.1. “exact” means $\mathcal{L} \rightarrow \mathcal{L}$; the charge is stated for the maps that preserve x_4 . P_b is the twisted Pin lift of the unit vector γ^b , which acts as the reflection of x_b , and P'_b is the other lift.

Map	Definition	Commuting	Anticommuting
C_0	$\Psi^*(x)$	exact, $Q \rightarrow -Q$	$-\mathcal{L}_{m,-\lambda}$, $Q \rightarrow Q$
C_8	$\gamma^8 \Psi^*(x)$	$-\mathcal{L}_{-m,-\lambda}$, $Q \rightarrow Q$	$\mathcal{L}_{-m,\lambda}$, $Q \rightarrow -Q$
chirality	$\gamma^8 \Psi(x)$	$-\mathcal{L}_{-m,-\lambda}$, $Q \rightarrow -Q$	$-\mathcal{L}_{-m,-\lambda}$, $Q \rightarrow -Q$
P_b , $b \leq 3$	$\gamma^b \Psi(R_b x)$	$\mathcal{L}_{-m,\lambda}$	$\mathcal{L}_{-m,\lambda}$
P'_b , $b \leq 3$	$\Gamma_{\{b\}^c} \Psi(R_b x)$	$-\mathcal{L}_{m,-\lambda}$	$-\mathcal{L}_{m,-\lambda}$
$C_0 P_b$	$\gamma^b \Psi^*(R_b x)$	$\mathcal{L}_{-m,\lambda}$	$-\mathcal{L}_{-m,-\lambda}$
$C_8 P_b$	$\Gamma_{\{b\}^c} \Psi^*(R_b x)$	$-\mathcal{L}_{m,-\lambda}$	exact, $Q \rightarrow -Q$
P_{123}	$\gamma^1 \gamma^2 \gamma^3 \Psi(R_{123} x)$	$\mathcal{L}_{-m,\lambda}$	$\mathcal{L}_{-m,\lambda}$
$C_8 P_{123}$	$\Gamma_{\{0,4,5,6,7\}} \Psi^*(R_{123} x)$	$-\mathcal{L}_{m,-\lambda}$	exact, $Q \rightarrow -Q$
P_{0123}	$C \Psi(R_{0123} x)$	exact, $Q \rightarrow Q$	exact, $Q \rightarrow Q$
$C_0 P_{0123}$	$C \Psi^*(R_{0123} x)$	exact, $Q \rightarrow -Q$	$-\mathcal{L}_{m,-\lambda}$
T	$\Gamma_{\{4\}^c} \Psi(R_4 x)$	exact	exact
T , antilinear	$\Gamma_{\{4\}^c} \Psi^*(R_4 x)$	exact	$-\mathcal{L}_{m,-\lambda}$
total inversion	$\gamma^8 \Psi(-x)$	exact	exact
CPT	$\gamma^8 \Psi^*(-x)$	exact	$-\mathcal{L}_{m,-\lambda}$
$CP_{123} T$	$\gamma^1 \gamma^2 \gamma^3 \gamma^4 \Psi^*(R_{1234} x)$	$-\mathcal{L}_{m,-\lambda}$	exact

Reading of the table:

- **Commuting field (dirac16complex00).** C is exact and reverses the charge. P and CP with one reflected space-like direction (or three) are not exact for $m \neq 0$; P is exact at $m = 0$. CP with an even number of reflected space-like directions, for example $C_0 P_{0123}$, is exact. T (linear and antilinear) and the full antilinear inversion CPT are exact.

- **Anticommuting field (dirac16complex).** No constant C is exact for $m \neq 0$: C_0 gives $-\mathcal{L}_{m,-\lambda}$, and the charge conjugation of unitary type, C_8 , gives $\mathcal{L}_{-m,\lambda}$, the theory with the opposite mass (C is exact at $m = 0$). CP with an odd number of reflected space-like directions is exact and reverses the charge: $C_8 P_b$ for $b = 0, 1, 2, 3$ and $C_8 P_{123}$ (in flat space and in gravitational fields with the reflection isometry, Section 5.3); $C_0 P_b$ is not exact. T is exact as a linear map; on the canonical structure it acts as an antilinear (antiunitary-type) automorphism (derived, Section 5.6). The antilinear total inversion is not exact, while $CP_{123}T$ is.

For both fields an exact symmetry exists that preserves x_4 and reverses Q : C_0 for the commuting field (in every gravitational field), $C_8 P_b$ for the anticommuting field (in flat space and in gravitational fields with the reflection isometry).

5.6 The quantized anticommuting field

The canonical anticommutator is $\{\Psi_a, \Psi_b^\dagger\} = B_{ab} \delta^7$ (flat space, Gaussian normal gauge). A map is compatible with it if the transformed operators satisfy the same relation. It is then a linear automorphism of the anticommutation relations (of unitary type) if the c-number B is reproduced, and it can only be an antilinear one (of antiunitary type) if $B^* = -B$ is reproduced, because an antiunitary operator complex-conjugates c-numbers. An automorphism of the anticommutation relations is not yet an operator on a given Fock space; whether it is implemented there is a separate question. From Lemma B, $MCM^{-1} = \sigma_R C$ and $M\gamma^4 M^{-1} = \varepsilon r_4 \gamma^4$, and $B^T = -B$, so

$$\begin{aligned} \text{linear } \Psi \rightarrow M\Psi : \quad & \{M\Psi, (M\Psi)^\dagger\} = MBM^\dagger = \sigma_R \varepsilon r_4 B, \\ \text{antilinear } \Psi \rightarrow M\Psi^{\dagger T} : \quad & \{M\Psi^{\dagger T}, (M\Psi^{\dagger T})^\dagger\} = MB^T M^\dagger = -\sigma_R \varepsilon r_4 B. \end{aligned}$$

For an exact symmetry of the anticommuting field ($\sigma_R \varepsilon = 1$ for linear, $\sigma_R = -1$ and $\varepsilon = 1$ for antilinear maps) both expressions equal $r_4 B$ (derived). Hence every exact symmetry is compatible with the canonical structure: as an automorphism of unitary type if it preserves x_4 and of antiunitary type if it reverses x_4 . The charge conjugation of unitary type is C_8 ($MB^T M^\dagger = +B$), while C_0 could only be of antiunitary type although it preserves x_4 ($MB^T M^\dagger = -B$). For CP , acting as a linear automorphism α of the field algebra, $\alpha(J^4) = (M\Psi^{\dagger T})^\dagger B M\Psi^{\dagger T}$; reordering the operator product produces the sign of Lemma C and a c-number, so the normal-ordered charge is reversed, $:Q: \rightarrow -:Q:$ (derived). Whether each map is also implemented by a unitary or antiunitary operator on the positive ($J = B$) Fock space of the good sector, and whether it leaves the vacuum invariant, is not decided here.

5.7 A remark on the slicing in signature (4,4)

The identity component $\text{Spin}_0(4,4)$, under which $\mathcal{L}$ is invariant, contains $\exp(\pi S^{45}) = \gamma^4 \gamma^5$ ($S^{45} = \frac{1}{2} \gamma^4 \gamma^5$ with $(S^{45})^2 = -\frac{1}{4}$), the rotation by π in the plane of the two time-like directions x_4, x_5 , and the product of four such rotations, γ^8 , the total inversion. Both are exact symmetries. For the classical fields both reverse x_4 and the x_4 -component of the current. In signature (4,4) the time orientation of the slicing $x_4 = \text{const}$ can therefore be reversed continuously, and the sign of the classical charge Q refers to a chosen slicing. This remark concerns the classical fields only. After quantization of the anticommuting field both maps have $MBM^\dagger = -B$, so by the

criterion of Section 5.6 they are of antiunitary type, like time reversal. For such an antilinear automorphism A with M real, $A J^4 A^{-1} = \Psi^\dagger M^T B^* M \Psi = -\Psi^\dagger M^T B M \Psi = +J^4$ at the reflected point, because $M^T B M = -B$ and $B^* = -B$: the quantum charge is preserved, not reversed (check `MA_M2_spin0ContainsChargeReversingTimeRotation` verifies the three matrix identities for both maps). These maps reverse x_4 ; they are not C or CP in Sakharov's sense and are not used in Section 9.

5.8 How M2 was verified

Wolfram: the intertwiner spaces of Lemma A are one-dimensional each (bases $1, \gamma^8, C, \gamma^8 C$, all commuting with every S^{ab}); the 256 monomials realise the 256 sign patterns bijectively, with the explicit null spaces; the statistics sign s is computed with an exact random Gaussian-rational Y ; all 512 pairs (M, R) satisfy the rules for ε, σ_R and κ ; for all 1024 maps per statistics, the Lagrangian density computed from the transformed jets (commuting symbols, respectively the Grassmann algebra) equals $\kappa \mathcal{L}_{\sigma\kappa m, \kappa\lambda}$ as given by the rule, and the eight current signs agree, with no mismatch; the frame form in the curved field G1 is checked for 31 frame reflections (21 at p1, 5 each at p2 and p3) for both statistics; and all 256 exact maps of the anticommuting field preserve the canonical structure as in Section 5.6.

Python (independent): the solution spaces of Lemma A, including the linear intertwiners 1 and γ^8 ; the internal maps C_0, C_8 and the chirality map on curved jets in `G_A`, agreeing with the hand derivation $(\kappa, \sigma, q) = (s, s, -s)$ for C_0 and $C_8 = C_0 \circ \text{chirality}$; the scope of CP in a gravitational field (Section 5.3: the internal parts of $C_8 P_b$ and $C_8 P_{123}$ at a fixed frame of `G_A`, the space-block determinant of the 128 frame reflections with s_R odd, and the non-exactness of their linear Pin lifts); named reflections on flat jets for both lifts; three generic, non-axis unit vectors $u = (2, 0, 0, 0, 1, 1, 1, 0)$ with $n(u) = +1$, $u = (1, 1, 0, 0, 1, 0, 1, 1)$ and $u = (0, 1, 0, 1, 1, 1, 1, 0)$ with $n(u) = -1$; reflections in the curved field `G_B`, which has reflection isometries; the rotation $\exp(\pi S^{45})$ and the total inversion of Section 5.7, including $MBM^\dagger = -B$, $M^T B M = -B$ and $M^T B^* M = +B$ for both; the complete character table of the 1024 maps per statistics (4 classes of 256, 256 exact symmetries, 64 exact charge-reversing symmetries without x_4 reversal), with multiplicativity of the characters checked on 400 random composites; the C and CP status; the canonical structure of the 256 exact maps of the anticommuting field (128 of unitary type and x_4 -preserving, 128 of antiunitary type and x_4 -reversing, none other); and C combined with spatial parities.

checks (Wolfram)

```
MA_M2_conjugationIntertwiners MA_M2_transposeIntertwiners
MA_M2_signPatternClassification
MA_M2_statisticsSign MA_M2_matrixClassification MA_M2_lagrangianFlatCommuting_all
MA_M2_lagrangianFlatGrassmann_all MA_M2_namedTransformations
MA_M2_frameLevelG1_commuting MA_M2_frameLevelG1_grassmann
MA_M2_symmetrySummaryAndChargeReversal MA_M2_canonicalStructure
```

checks (Python)

```
MA_M2_chargeConjugationSolutionSpaces MA_M2_psiBarTransposeFormsReduceToPsiStarForms
MA_M2_internalMapsCurvedJets MA_M2_internalMapsAsDerived
MA_M2_cpScopeInCurvedFields MA_M2_namedReflectionsFlatJets
MA_M2_genericUnitVectorReflections MA_M2_reflectionsInCurvedFieldWithIsometry
MA_M2_spin0ContainsChargeReversingTimeRotation MA_M2_discreteGroupCharacterTable
MA_M2_characterHomomorphism MA_M2_C_and_CP_status
MA_M2_canonicalStructureOfExactGrassmannSymmetries
```

6 Theorem M3: the charge-violating terms allowed by the symmetry

Sakharov's first condition needs an interaction that changes the charge. The simplest candidates are “Majorana-type” terms built with Ψ^T instead of $\Psi^\dagger$: under $\Psi \rightarrow e^{i\alpha}\Psi$ they are multiplied by $e^{2i\alpha}$ (U(1) charge 2), so they violate the U(1) of M1. Theorem M3 classifies the ones that the local Lorentz symmetry allows. It is a classification, not a claim that any such term is present in the theory or natural.

6.1 Statement

Theorem M3.

1. (invariant forms) The constant matrices M with $(S^{ab})^T M + M S^{ab} = 0$ for all 28 generators, that is, with $\Psi^T M \Psi$ invariant under $\text{Spin}_0(4, 4)$, are exactly $M = C(\alpha P_- + \beta P_+)$ with complex α, β . The space has dimension 2, all its elements are symmetric, and it contains no nonzero antisymmetric matrix.
2. (the full group) The elements of $\text{Spin}(4, 4)$ of spinor norm -1 , such as $g = \gamma^a \gamma^b$ with a space-like and b time-like, act by $g^T (C P_\pm) g = -C P_\pm$. No nonzero form is strictly invariant under the full $\text{Spin}(4, 4)$; both basis forms carry the spinor-norm character. (Stage 1, Result 10.3, calls $C P_\pm$ the $\text{Spin}(4, 4)$ -invariant bilinear forms. There, invariance means invariance under the 28 generators, that is under $\text{Spin}_0(4, 4)$, which is item 1; item 2 refines that statement: under the component of spinor norm -1 both forms change sign.)
3. (Pin characters) For every unit vector $u = u_a \gamma^a$, $u^2 = n(u) = \pm 1$: $u^T C u = -n(u) C$ and $u^T (C \gamma^8) u = +n(u) C \gamma^8$. Among the Spin_0 -invariant forms, the trivial character $\chi = 1$ and the determinant character $\chi(u) = -1$ admit no nonzero covariant form; C carries $\chi(u) = -n(u)$ (the character of $\bar{\Psi} \Psi$), and $C \gamma^8$ carries $\chi(u) = +n(u)$. The chiral forms $C P_\pm$ are Spin_0 -invariant but not Pin-covariant, because u exchanges the two chiralities.
4. (derivative terms) The derivative bilinears $\Psi^T M \gamma^a \partial_a \Psi$ that are Spin_0 -invariant have M in the same two-dimensional space. $M \gamma^a$ is symmetric for every a exactly when $M \propto C \gamma^8$, and antisymmetric for every a exactly when $M \propto C$.
5. (what survives) For **anticommuting** components every invariant mass-type term $\Psi^T M \Psi$ vanishes identically: no Majorana mass term exists. The derivative term $\sqrt{|g|} \Psi^T C \gamma^\mu D_\mu \Psi$ is a total divergence (identically vanishing Euler-Lagrange equations), while $\sqrt{|g|} \Psi^T C \gamma^8 \gamma^\mu D_\mu \Psi$ survives, with flat-space Euler-Lagrange expression $2C \gamma^8 \gamma^a \partial_a \Psi$. For **commuting** components the two chiral mass terms $\Psi^T C P_\pm \Psi$ survive (equivalently $\Psi^T C \Psi$ and $\Psi^T C \gamma^8 \Psi$), and so does the derivative term $\sqrt{|g|} \Psi^T C \gamma^\mu D_\mu \Psi$, which is the form of the notebook Lagrangian $\text{Lg}[]$; $\sqrt{|g|} \Psi^T C \gamma^8 \gamma^\mu D_\mu \Psi$ is a total divergence.
6. (charge) Every term of items 1 to 5 has U(1) charge 2, and none of them is contained in $\mathcal{L}$.

6.2 Proof

Item 1. From $(CS^{ab})^T = -CS^{ab}$ (Stage 1, Result 3.5) and $C^T = C$ one gets $(S^{ab})^T = -CS^{ab}C^{-1}$. The condition becomes $-CS^{ab}C^{-1}M + MS^{ab} = 0$, that is, $S^{ab}(C^{-1}M) = (C^{-1}M)S^{ab}$: the matrix $C^{-1}M$ lies in the commutant of the 28 generators. That commutant is spanned by P_- and P_+ (Stage 1, Result 4.2, dimension 2). So $M = C(\alpha P_- + \beta P_+)$. Since γ^8 is symmetric and commutes with C , $(CP_{\pm})^T = P_{\pm}C = CP_{\pm}$: both are symmetric, and a nonzero antisymmetric solution cannot exist.

Item 2. By the proof of Lemma B, $(\gamma^a)^T C \gamma^a = -C$ for $a \leq 3$ and $+C$ for $a \geq 4$; since γ^8 commutes with C and anticommutes with γ^a , $(\gamma^a)^T C \gamma^8 \gamma^a = -(\gamma^a)^T C \gamma^a \gamma^8$, which is $+C\gamma^8$ for $a \leq 3$ and $-C\gamma^8$ for $a \geq 4$. For $g = \gamma^a \gamma^b$ with $a \leq 3 < b$ this gives $g^T C g = -C$ and $g^T (C\gamma^8) g = -C\gamma^8$, hence $g^T (CP_{\pm}) g = -CP_{\pm}$. The elements of spinor norm -1 form the coset $g \text{Spin}_0(4, 4)$, and the forms of item 1 are invariant under $\text{Spin}_0(4, 4)$, so every such element changes the sign of every form of item 1, and no nonzero form is invariant under both components.

Item 3. $(C\gamma^a)^T = -C\gamma^a$ gives $(\gamma^a)^T C = -C\gamma^a$, hence $u^T C u = -C u u = -n(u)C$. With $\gamma^8 u = -u\gamma^8$, $u^T C \gamma^8 u = -C u \gamma^8 u = C \gamma^8 u^2 = n(u)C\gamma^8$. A form $\alpha CP_- + \beta CP_+$ is covariant for a character only if it is proportional to C or to $C\gamma^8$, because $u^T (CP_{\pm}) u = -n(u)CP_{\mp}$ exchanges the two chiral pieces.

Item 4. $M\gamma^a$ symmetric for all a : $(C\gamma^8 \gamma^a)^T = (\gamma^a)^T \gamma^8 C = (\gamma^a)^T C \gamma^8 = -C\gamma^a \gamma^8 = C\gamma^8 \gamma^a$. $M\gamma^a$ antisymmetric for all a : $(C\gamma^a)^T = -C\gamma^a$ (expression [1] of Stage 1). The combinations $\alpha CP_- + \beta CP_+$ with both α and β nonzero and $\alpha \neq \pm\beta$ have neither property; the exact solutions are $\beta = -\alpha$ (symmetric) and $\beta = \alpha$ (antisymmetric).

Item 5. For anticommuting components $\Psi_i \Psi_j = -\Psi_j \Psi_i$, so $\Psi^T M \Psi = \Psi^T M_A \Psi$ with $M_A = \frac{1}{2}(M - M^T)$. The map $M \mapsto (S^{ab})^T M + MS^{ab}$ sends symmetric matrices to symmetric ones and antisymmetric matrices to antisymmetric ones, so the invariance of $\Psi^T M_A \Psi$ is the condition of item 1 for M_A alone. By item 1, M_A is then symmetric and antisymmetric at once, hence zero: every invariant mass-type term vanishes. For commuting components only the symmetric part $M_S = \frac{1}{2}(M + M^T)$ counts, and $CP_{\pm}$ are symmetric and nonzero. For derivative terms, if A is antisymmetric and the components anticommute, then $\partial_a(\Psi^T A \Psi) = 2\Psi^T A \partial_a \Psi$, so $\Psi^T C \gamma^a \partial_a \Psi = \frac{1}{2}\partial_a(\Psi^T C \gamma^a \Psi)$ is a total derivative. In curved space, with $\sqrt{|g|}$ and the canonical connection, this is the Stage-1 theorem that $\text{Lg}[\Psi]$ is a pure divergence for a Grassmann field (Stage 1, Section 6). If instead X is symmetric, varying $\Psi^T X \partial_a \Psi$ and integrating by parts gives the Euler-Lagrange expression $2X \partial_a \Psi \neq 0$. For commuting components the roles of symmetric and antisymmetric are exchanged.

Item 6. $\Psi^T \rightarrow e^{i\alpha} \Psi^T$ and $\Psi \rightarrow e^{i\alpha} \Psi$, so every term $\Psi^T X \Psi$ is multiplied by $e^{2i\alpha}$. $\mathcal{L}$ contains only $\Psi^\dagger \dots \Psi$ bilinears and functions of S . $\square$

Pin characters of the Majorana-type terms. For $\Psi \rightarrow u\Psi$ with $u = \gamma^b$ ($n = \eta_{bb}$): $\Psi^T C \Psi$ picks up $-n$ and $\Psi^T C \gamma^8 \Psi$ picks up $+n$; the kinetic term $\Psi^T C \gamma^a \partial_a \Psi$ picks up $+n$ with the twisted lift and $-n$ with the untwisted lift, and $\Psi^T C \gamma^8 \gamma^a \partial_a \Psi$ picks up $-n$ (twisted) and $+n$ (untwisted).

The notebook Lagrangian. $\text{Lg}[\Psi]$ uses $\text{Transpose}[\text{Psi16}]$, not $\text{ConjugateTranspose}$: its kinetic matrix $\sigma_{16} T 16^a = C\gamma^a$ and its mass matrix $\sigma_{16} = C$ belong to the family of item 1 with the Pin character $-n(u)$. For a complex commuting field it has $\text{U}(1)$ charge 2; for a real commuting field (the Stage-5 real restriction) it is non-trivial; for a Grassmann field it is a total derivative with a vanishing mass term (Stage 1). `dirac16complex00` uses the charge-0 Lagrangian $\mathcal{L}$ instead.

6.3 Quartic examples (not a classification)

For anticommuting components invariant charge-violating terms of higher order exist. Since C is an invariant form and $\gamma^a\gamma^b$ transforms as a tensor, the bilinears $T^{ab} = \Psi^T C \gamma^a \gamma^b \Psi$ ($a \neq b$, antisymmetric matrices, hence non-vanishing for Grassmann components) form an antisymmetric tensor, and full contractions are Spin_0 -invariant:

$$Q_4 = \sum_{a < b} \eta_{aa} \eta_{bb} (\Psi^T C \gamma^a \gamma^b \Psi)^2 \quad (40 \text{ monomials, charge } 4),$$

$$Q_2 = \sum_{a < b} \eta_{aa} \eta_{bb} (\Psi^T C \gamma^a \gamma^b \Psi) (\Psi^\dagger C \gamma^a \gamma^b \Psi) \quad (160 \text{ monomials, charge } 2).$$

Q_4 equals exactly 8 times the chiral form $\sum_{a < b} \eta_{aa} \eta_{bb} (\Psi^T C P_- S^{ab} \Psi) (\Psi^T C P_+ S^{ab} \Psi)$ of the Wolfram package, which is invariant under all 28 generators (a single term is not, the control), while the purely chiral quartics vanish. The analogous contraction with one factor $\Psi^T C \gamma^8 \gamma^a \gamma^b \Psi$ vanishes, and for commuting components all three contractions vanish because $C \gamma^a \gamma^b$ is antisymmetric. These are examples only: higher-order charge-violating terms are not classified, and nothing here says that such a term is present, natural, or sufficient for generating an asymmetry.

6.4 How M3 was verified

Wolfram: the null space of the 28 conditions has dimension 2 with basis CP_- , CP_+ , symmetric subspace of dimension 2 and antisymmetric subspace of dimension 0; the character table over the four characters (dimensions 0, 1, 1, 0 for the trivial, $-n$, $+n$ and determinant characters), confirmed on two non-basis unit vectors ($\gamma^0 + \gamma^1 + \gamma^4$ with $n = +1$, $\gamma^0 + \gamma^4 + \gamma^5$ with $n = -1$) and for the spinor-norm -1 component; the kinetic forms (the same space; $\beta = -\alpha$ symmetric, $\beta = \alpha$ antisymmetric); survival: all four mass-type forms vanish for Grassmann components (a control with an antisymmetric matrix has 8 nonzero monomials), the commuting forms have ranks 8, 8, 16 and 16 for CP_- , CP_+ , C and $C\gamma^8$; the kinetic Euler-Lagrange expressions ($2C\gamma^8\gamma^a\partial_a\Psi$ for Grassmann, $2C\gamma^a\partial_a\Psi$ for commuting, the other form a total derivative); the U(1) charge 2; the Pin characters of the four terms; the notebook Lg[] identification; the quartic Q_4 .

Python (independent): the null space of $S^{abT}M + MS^{ab} = 0$ as an exact integer basis equal to $\frac{1}{2}(C\gamma^8 - C)$ and $\frac{1}{2}(C\gamma^8 + C)$; symmetric dimension 2, antisymmetric 0; strictly invariant under the full $\text{Spin}(4,4)$: 0, covariant with the spinor-norm character: 2; the Pin characters of C and $C\gamma^8$; survival with monomial counts ($\Psi^T C \Psi$ and $\Psi^T C \gamma^8 \Psi$: 0 monomials for Grassmann, 8 each for commuting, while $\Psi^\dagger C \Psi$ and $\Psi^\dagger C \gamma^8 \Psi$ have 16 monomials and charge 0); the derivative terms in the curved field G_A ($\Psi^T C \gamma^8 \gamma^\mu D_\mu \Psi$ for Grassmann and $\Psi^T C \gamma^\mu D_\mu \Psi$ for commuting components with 16 nonzero Euler-Lagrange components each, the other two total divergences); the quartics Q_4 (40 monomials, charge 4) and Q_2 (160 monomials, charge 2), and the relation $Q_4 = 8 \times$ the Wolfram chiral form.

checks (Wolfram)

```
MA_M3_spinInvariantForms MA_M3_pinCharacterForms MA_M3_kineticInvariantForms
MA_M3_grassmannSurvival MA_M3_commutingSurvival MA_M3_u1Charge
MA_M3_pinCharactersMajoranaTerms MA_M3_notebookLgIsMajoranaType
MA_M3_extraGrassmannQuarticCharge4
```

checks (Python)

MA_M3_invariantFormsSpan_C_Cgamma8 MA_M3_allInvariantFormsSymmetric
MA_M3_fullSpinCharacter MA_M3_pinCharacters MA_M3_derivativeBilinearClassification
MA_M3_massTypeSurvivalAndCharge MA_M3_derivativeTypeSurvivalCurved
MA_M3_quarticChargeViolatingExamplesGrassmann MA_M3

7 Theorem M4: the chirality pair of universes

7.1 Field level

Theorem M4 (field level; Stage-5 theorem T1). For every vielbein, every m and λ , both statistics and every configuration Ψ (off shell), with $\Psi_- := \gamma^8 \Psi$:

$$\begin{aligned}\bar{\Psi}_- &= \bar{\Psi} \gamma^8, & S[\Psi_-] &= S[\Psi], & K[\Psi_-] &= -K[\Psi], & j^\mu[\Psi_-] &= -j^\mu[\Psi], \\ \mathcal{L}_{m,\lambda}[\gamma^8 \Psi] &= -\mathcal{L}_{-m,-\lambda}[\Psi], & E_{-m,-\lambda}[\gamma^8 \Psi] &= -\gamma^8 E_{m,\lambda}[\Psi], \\ T_{\mu\nu}[\gamma^8 \Psi; -m, -\lambda] &= -T_{\mu\nu}[\Psi; m, \lambda].\end{aligned}$$

Hence Ψ solves the field equations with (m, λ) if and only if $\gamma^8 \Psi$ solves them with $(-m, -\lambda)$.

Proof. γ^8 is Hermitian, $(\gamma^8)^2 = 1$, it anticommutes with every γ^a , and it commutes with C and with every S^{ab} , hence with every Ω_μ and with D_μ . Therefore $\bar{\Psi}_- = \Psi^\dagger \gamma^8 C = \bar{\Psi} \gamma^8$. A bilinear $\bar{\Psi} X \Psi$ goes to $\bar{\Psi} \gamma^8 X \gamma^8 \Psi$, which is $-\bar{\Psi} X \Psi$ when X contains one gamma matrix (the current, the kinetic term, the kinetic part of $T_{\mu\nu}$) and $+\bar{\Psi} X \Psi$ for $X = 1$ (the scalar S). Consequently

$$\mathcal{L}_{m,\lambda}[\gamma^8 \Psi] = \sqrt{|g|}[-K - mS - \frac{\lambda}{2} S^2] = -\sqrt{|g|}[K - (-m)S - \frac{(-\lambda)}{2} S^2] = -\mathcal{L}_{-m,-\lambda}[\Psi].$$

For the field equation, $\gamma^\mu D_\mu(\gamma^8 \Psi) = -\gamma^8 \gamma^\mu D_\mu \Psi$ and $(m + \lambda S)\gamma^8 \Psi = \gamma^8(m + \lambda S)\Psi$, so $E_{-m,-\lambda}[\gamma^8 \Psi] = -\gamma^8 \gamma^\mu D_\mu \Psi + \gamma^8(m + \lambda S)\Psi = -\gamma^8 E_{m,\lambda}[\Psi]$. $T_{\mu\nu} = -\frac{1}{4}[\text{kinetic bilinears}] + g_{\mu\nu} \mathcal{L}_s$ (Stage 1, Section 9): the kinetic bilinears change sign, and $\mathcal{L}_s[\gamma^8 \Psi; -m, -\lambda] = -\mathcal{L}_s[\Psi; m, \lambda]$ by the Lagrangian identity with (m, λ) replaced by $(-m, -\lambda)$. The map is linear and keeps every $\Psi^\dagger$ to the left of every Ψ , so the proof holds for both statistics. The even potential forces $\lambda \rightarrow -\lambda$: at fixed λ one gets $\mathcal{L}_{m,\lambda}[\gamma^8 \Psi] + \mathcal{L}_{-m,\lambda}[\Psi] = -\lambda \sqrt{|g|} S^2 \neq 0$ (CONTRACT erratum E2). $\square$

Corollary M4.1 (the pair, classical fields). Let Ψ_+ be a configuration of a classical field with the Lagrangian $\mathcal{L}_{m,\lambda}$, and let Ψ_- be the configuration $\gamma^8 \Psi_+$ of a second, independent classical field with the Lagrangian $\mathcal{L}_{-m,-\lambda}$ (equivalently, one whose action on the paired configuration is $S_{-m,-\lambda}[\gamma^8 \Psi_+] = -S_{m,\lambda}[\Psi_+]$), in the same gravitational field. Then, at every point and every x_4 (in particular at $x_4 = 0$),

$$j_+^\mu + j_-^\mu = 0, \quad Q_+ + Q_- = 0, \quad T_{\mu\nu}^{\text{pair}} = T_{\mu\nu}[\Psi_+; m, \lambda] + T_{\mu\nu}[\Psi_-; -m, -\lambda] = 0.$$

Here Q_- is the Noether charge of the second field in its own right (by M1, item 2, with $(-m, -\lambda)$ its Noether current is $i\sqrt{|g|} j^\mu[\Psi_-]$), and $T_{\mu\nu}[\Psi_-; -m, -\lambda]$ is its own metric energy-momentum tensor; in particular its classical energy is minus that of Ψ_+ . By Theorem M4 the second field solves its field equations exactly when the first one does. The pair carries no net charge and no net energy, momentum or stress, so the Einstein (or Einstein-Lovelock) equations with the pair as source are the source-free equations. This is a statement about classical fields, conservation laws and constraints.

The correlation $\Psi_- = \gamma^8 \Psi_+$ between the two fields is a property of the chosen configurations, not a consequence of any dynamics, and nothing here is a computed creation rate or amplitude. The quantum level is different (Section 7.3): there no reading of the pair gives a cancellation between two independent universes.

7.2 One-particle (Krein) level

In the good sector (flat space, no momenta along x_5, x_6, x_7) the mode Hamiltonian is $h_k(m) = -im\gamma^4 - \gamma^4 \sum_{j \neq 4} k_j \gamma^j$, Hermitian, commuting with B , with $h_k(m)^2 = (m^2 + k^2) 1$ (Stage 1, Section 10.6). Since $\gamma^8 \gamma^4 \gamma^8 = -\gamma^4$ and $\gamma^8 \gamma^4 \gamma^j \gamma^8 = \gamma^4 \gamma^j$,

$$\gamma^8 h_k(m) \gamma^8 = h_k(-m), \quad \gamma^8 B \gamma^8 = -B.$$

So γ^8 maps the positive-energy (negative-energy) eigenspace of $h_k(m)$ onto the positive-energy (negative-energy) eigenspace of $h_k(-m)$ with the same energy, it preserves the Hilbert norm $u^\dagger u$, and it reverses the Krein norm $u^\dagger B u$. At rest the B -form on each positive-energy space has signature (4,4), and the Gram matrix of the image is minus the original one.

Which charge and energy an image mode carries depends on the canonical structure given to the image field. With the Stage-1 expectation rule $\langle \Psi^\dagger X \Psi \rangle = u^\dagger B X u$ for a Hilbert-normalised positive-energy mode u , the energy is $X = B h$, the charge $X = B$ and the scalar density $X = C$. For the image field $\Psi_- = \gamma^8 \Psi$ the canonical anticommutator is $\gamma^8 B \gamma^8 = -B$ and the rule reads $u_-^\dagger (-B) X u_-$ with $u_- = \gamma^8 u$. For an independently quantised $-m$ theory with its own positive structure the rule is $u_-^\dagger B X u_-$. The table gives (E, Q, S) per quantum for three exact positive-energy modes u with mass m and momentum $k = (k_0, k_1, k_2, k_3)$: in the $+m$ universe; the formulas $H[\Psi_-; -m]$, $Q[\Psi_-]$ and $S[\Psi_-]$ of the Lagrangian $\mathcal{L}_{-m, -\lambda}$ evaluated on the image field with its metric $-B$; and the independently quantised field (metric $+B$).

m, k	$+m$ universe	$\mathcal{L}_{-m, -\lambda}$ formulas, im- age, $-B$	Independent, $+B$
1, (1, 1, 2, 3)	(4, 1, 1/4)	(-4, -1, 1/4)	(4, 1, -1/4)
3, (1, 1, 1, 2)	(4, 1, 3/4)	(-4, -1, 3/4)	(4, 1, -3/4)
3/5, (4/5, 0, 0, 0)	(1, 1, 3/5)	(-1, -1, 3/5)	(1, 1, -3/5)

In general (derived from the two matrix identities above): the $\mathcal{L}_{-m, -\lambda}$ formulas evaluated on the image field give $(E, Q, S) \rightarrow (-E, -Q, S)$ per quantum, and the independently quantised field has $(E, Q, S) \rightarrow (E, Q, -S)$ per quantum. The third column is not the energy and charge of the image field itself. With the anticommutator $-B$ the operator $H[\Psi_-; -m]$ generates the reversed x_4 -evolution and $Q[\Psi_-]$ the inverse phase, so the image field's own generators are $-H[\Psi_-; -m]$ and $-Q[\Psi_-]$, with $(+E, +Q)$ per quantum, as in the $+m$ universe (Section 7.3; the matrix identities $(-B)(B h_k(-m)) = -h_k(-m)$, $(-B)B = -1$ and $\gamma^8 B h_k(-m) \gamma^8 = -B h_k(m)$ are part of MA_M4_kreinOneParticle).

7.3 Fock level: the image field and an independently quantised field

What is proved here and what is cited. The matrix identities behind the Fock-level statements are proved here by both implementations (MA_M4_kreinOneParticle, MA_M4_kreinModeFacts). The canonical structure of the image field is proved here by the Python implementation only, in an exact Fock model (check MA_M4_imageFieldFockModel). Its four modes v_n of the good sector, with $m = 1$ at rest and at $k = (1, 1, 2, 3)$, are joint eigenvectors of $h_k(m)$ (energy $\epsilon_n = \pm E$) and of B (Krein sign $\beta_n = \pm 1$). They are realised on the 16-dimensional Fock space of four fermionic modes with exact Gaussian-rational matrices, with the Krein anticommutators $\{b_n, b_n^K\} = \beta_n$ and without normal ordering, which changes H and Q by constants and leaves every commutator below unchanged. The Stage-5 Fock-level result PAIR_T1krein is cited, not re-proved. It uses a Fock model of four rest-sector modes with $m = 1$, the Krein anticommutators $\{b_n, b_n^K\} = \beta_n$, the sea formed by the two negative-energy modes and normal ordering as subtraction of the sea value. The Stage-5 Wolfram pairing report is present, all of its 141 checks are true (among them the twelve PAIR_T1krein checks), and the independent Stage-5 Python pairing report has 172 of 172 checks true (among them S5_T1krein). The Stage-5 documents and the Stage-5 gate are not complete, so both matter-antimatter reports record this citation as PROVISIONAL, open until the Stage-5 gate passes (Section 11.4).

The image field. The image field $\Psi_- = \gamma^8 \Psi$ on the same Fock space and in the same state has the anticommutator $-B$ on its modes. The operator identities $H[\Psi_-; -m] = -H[\Psi; m]$, $Q[\Psi_-] = -Q[\Psi]$ and $S[\Psi_-] = S[\Psi]$ hold, with $H = \Psi^K B h \Psi$ and $Q = \Psi^K B \Psi$ (Stage 5; re-proved in MA_M4_imageFieldFockModel). Normal ordering commutes with the map, and $:T_{\mu\nu}[\gamma^8 \Psi; -m, -\lambda] := - :T_{\mu\nu}[\Psi; m, \lambda] :$ as operators (Stage 5, cited). Stage 5 describes the image as a universe with the Krein metric $-B$, energy $-|\epsilon|$ and charge -1 per quantum. These numbers are the expectation values of the $\mathcal{L}_{-m, -\lambda}$ formulas; they are not the image field's own Hamiltonian and charge. For operators with $\{\Psi_a, \Psi_b^K\} = G_{ab}$ one has $[\Psi, \Psi^K X \Psi] = G X \Psi$, and with $G = -B$

$$[\Psi_-, H[\Psi_-; -m]] = -h(-m) \Psi_-, \quad [\Psi_-, Q[\Psi_-]] = -\Psi_-,$$

while the image field evolves by $i\partial_4 \Psi_- = \gamma^8 h(m) \gamma^8 \Psi_- = h(-m) \Psi_-$. So $H[\Psi_-; -m]$ generates the reversed x_4 -evolution and $Q[\Psi_-]$ the inverse phase, and the image field's own x_4 -generator and U(1) generator are

$$-H[\Psi_-; -m] = +H_+, \quad -Q[\Psi_-] = +Q_+.$$

Its canonical Lagrangian is $-\mathcal{L}_{-m, -\lambda} = \mathcal{L}_{m, \lambda}[\gamma^8 \cdot]$ (Stage 5). Its Noether current is $-i\sqrt{|g|} j^\mu[\Psi_-] = +i\sqrt{|g|} j^\mu[\Psi_+]$ (M1, item 2), and its metric energy-momentum tensor is $-T_{\mu\nu}[\Psi_-; -m, -\lambda] = +T_{\mu\nu}[\Psi_+; m, \lambda]$ (M4) (derived). Relative to its own canonical structure the image field therefore has the energy $+H_+$, the charge $+Q_+$ and the gravitational source $+T_+$. It is the same quantum system as Ψ_+ written in other variables: the same operators, the same Fock space and the same state. The identities $Q_+ + Q[\Psi_-] = 0$ and $H_+ + H[\Psi_-; -m] = 0$ have the form $X + (-X) = 0$ and hold in every state, including every state with $Q_+ \neq 0$ (in the Fock model Q_+ takes the values 0 to 4). They are not a compensation by a second universe.

An independently quantised field. An independently quantised theory $\mathcal{L}_{-m, -\lambda}$ with its own anticommutator $+B$ and the positive ($J = B$) structure has the modes $w_n = \gamma^8 u_n$ (same ϵ , Krein sign $-\beta_n$); a quantum in w_n has energy $+|\epsilon|$, charge $+1$ and scalar density $-u^\dagger B C u$ (Stage 5).

In the state with the same mode occupations its free Hamiltonian and its charge equal those of the $+m$ universe, and its scalar density is reversed: $H' = H_+$, $Q' = Q_+$ and $S' = -S_+$ as operators (MA_M4_imageFieldFockModel). The kinetic and mass energies therefore add instead of cancelling. The interaction energy changes sign: with $(-m, -\lambda)$ the potential of the second field is $-\frac{\lambda}{2}S'^2$, and $S'^2 = S_+^2$ (checked in the Fock model), so it equals $-\frac{\lambda}{2}S_+^2$. Equivalently, the Wick form $\langle U \rangle = \frac{\lambda}{2}[\text{Tr}(M\rho)^2 - \text{Tr}(M\rho M\rho)]$ of Stage 5 is even in the scalar-density matrix and odd in λ (derived). Equal energies hold for $\lambda = 0$ (the free Stage-5 Fock model) and for the choice $(-m, +\lambda)$ of the Stage-5 theorem T2, which Stage 5 calls the T2-type pairing. For the $(-m, -\lambda)$ partner of M4 the kinetic and mass energies add and the interaction energies cancel.

How particles and antiparticles map (derived from these operator identities). In the $+m$ universe a particle has $(E, Q) = (|\epsilon|, +1)$ and an antiparticle (a hole in the sea) has $(|\epsilon|, -1)$. The image field describes the same particles and antiparticles, with the same energies and charges when these are measured with its own generators $-H[\Psi_-; -m]$ and $-Q[\Psi_-]$; the $\mathcal{L}_{-m, -\lambda}$ formulas evaluated on the same states give $(-|\epsilon|, -1)$ and $(-|\epsilon|, +1)$. In the independently quantised $-M$ universe particles and antiparticles have the same energies and charges as in the $+M$ universe and the opposite scalar density. For the positive- and negative-norm modes: a mode u_n with Krein sign $\beta_n = u_n^\dagger B u_n$ goes to $w_n = \gamma^8 u_n$ with $w_n^\dagger B w_n = -\beta_n$, because $\gamma^8 B \gamma^8 = -B$. Measured with the image field's own metric $-B$ the sign is β_n again, so the image field reproduces the occupation pattern of the $+M$ universe; measured with $+B$ (independent quantization) every positive-norm mode becomes a negative-norm mode and conversely (derived from the matrix identity and the Stage-5 mode list; the sign flip is also checked in MA_M4_imageFieldFockModel).

Consequence for the pair. At the quantum level no reading gives a cancellation between two independent, consistently quantised universes. In the image-field reading the $-M$ member has no degrees of freedom of its own. Counted with its own energy-momentum tensor it would source $+T_+$, so a pair counted as two sources would source $2T_+$, not 0; since it is the same system, it adds nothing. In the independent reading the kinetic and mass energies add, and the charges cancel only if the $-M$ universe is in a state with the opposite charge, which is an additional assumption about its state. The cancellation of Corollary M4.1 is therefore a statement about classical fields only.

7.4 How M4 was verified

Wolfram: the matrix facts (the current matrices are odd under γ^8 , the mass matrix is even, $\bar{\Psi} \rightarrow \bar{\Psi}\gamma^8$, $B \rightarrow -B$); the current flip in G1 for Grassmann components at three points; for commuting components at the three G1 points the jets of $T_{\mu\nu}^{\text{pair}}$ and of the pair current vanish while the individual ones do not, and the image of an on-shell solution solves the $(-m, -\lambda)$ equations; for Grassmann components all 64 components of $T_{\mu\nu}^{\text{pair}}$ vanish at p1 (64 nonzero components individually, symmetric); the one-particle facts of Section 7.2, including the canonical generators of the image field at the matrix level. The verifier also reads the Stage-5 pairing files and records their hashes and the status of the citation; this is not a check.

Python (independent): the γ^8 matrix facts, including $(\gamma^8)^T C \gamma^a \gamma^8 = -C \gamma^a$ and commutation with Ω_μ in G_A; in G_A for both statistics the current flip, S invariant, $K \rightarrow -K$, $\mathcal{L}_{m, \lambda}[\gamma^8 \Psi] = -\mathcal{L}_{-m, -\lambda}[\Psi]$, the Euler-Lagrange pairing and the pairing of all 36 independent components of $T_{\mu\nu}$ (36 nonzero), so the pair totals vanish identically as polynomial identities; the three Krein mode samples of the table in Section 7.2; the exact Fock model of Section 7.3 (the anticommutators B and $-B$, the generators of the image field, the operator identities, and the independently quantised

field with $H' = H_+$, $Q' = Q_+$, $S' = -S_+$ and $S'^2 = S_+^2$); the citation of the Stage-5 reports with all twelve PAIR_T1krein checks and all fourteen S5_T1krein checks true.

```
checks (Wolfram)
  MA_M4_currentFlipMatrix MA_M4_currentFlipG1_grassmann
  MA_M4_pairEMTAndCurrentG1_commuting
  MA_M4_pairEMTG1_grassmann MA_M4_kreinOneParticle
checks (Python)
  MA_M4_gamma8MatrixFacts MA_M4_gamma8ChargeFlipCurved MA_M4_gamma8EulerLagrangePairing
  MA_M4_gamma8EMTPairing MA_M4_kreinModeFacts MA_M4_imageFieldFockModel MA_M4
checks (Stage-5 Wolfram pairing report, cited)
  PAIR_T1krein_restHamiltonians PAIR_T1krein_modes PAIR_T1krein_fieldCAR
  PAIR_T1krein_statesNormalised PAIR_T1krein_expectationRule
  PAIR_T1krein_positiveExcitations PAIR_T1krein_imageAnticommutatorMinusB
  PAIR_T1krein_imageOperatorIdentities PAIR_T1krein_imageExpectationValues
  PAIR_T1krein_minusMModes PAIR_T1krein_independentCARPlusB
  PAIR_T1krein_independentExpectationValues
checks (Stage-5 Python pairing report, cited)
  S5_T1krein_restHamiltonians S5_T1krein_modes S5_T1krein_fieldCAR
  S5_T1krein_bilinearFromFields S5_T1krein_statesNormalised
  S5_T1krein_expectationRule S5_T1krein_positiveExcitations
  S5_T1krein_imageAnticommutatorMinusB S5_T1krein_imageOperatorIdentities
  S5_T1krein_imageExpectationValues S5_T1krein_minusMModes
  S5_T1krein_independentCARPlusB S5_T1krein_independentExpectationValues S5_T1krein
```

8 M5: the conditional scenario, stated as a hypothesis

8.1 The hypotheses

The following three statements are **hypotheses**. None of them is derived anywhere in this repository.

- **H1 (hypothesis, not derived).** Our universe is one member of a chirality pair created together: two independent classical fields in the same gravitational field, Ψ_+ with the Lagrangian $\mathcal{L}_{m,\lambda}$ and Ψ_- with the Lagrangian $\mathcal{L}_{-m,-\lambda}$, in the correlated configuration $\Psi_- = \gamma^8 \Psi_+$ of Corollary M4.1. The correlation between the two fields is part of the assumption. H1 is a statement about classical fields; it has no quantum counterpart with two independent universes (Section 7.3). No creation process, rate or amplitude is computed anywhere in this repository.
- **H2 (hypothesis, not derived).** The creation assigns $Q_+ = -Q_- \neq 0$. Given H1, the relation $Q_- = -Q_+$ follows from M4; the content of H2 is $Q_+ \neq 0$, and nothing here computes the value or the sign of Q_+ .
- **H3 (hypothesis, not derivable within the theory).** The dirac16complex U(1) charge is identified with baryon number B (or with $B - L$). The theory contains no Standard-Model baryons, quarks or leptons, so this identification cannot be derived from it.

8.2 The implication and its proof

Proposition M5. If H1, H2 and H3 hold, then, for the two classical fields of H1, $Q_+(x_4) + Q_-(x_4) = 0$ at every x_4 , each of Q_+ and Q_- is separately conserved, and the baryon excess $B_+ = Q_+ \neq 0$ of one member is balanced by $B_- = -Q_+$ of the other. The pair also carries $T_{\mu\nu}^{\text{pair}} = 0$ (classical bilinears). The balance is put in by the correlated configuration assumed in H1; it is not a compensation mechanism, and it has no quantum counterpart with two independent universes (Section 7.3).

Proof. By M4, $\sqrt{|g|} j^{x_4} [\gamma^8 \Psi_+] = -\sqrt{|g|} j^{x_4} [\Psi_+]$ at every point, and the Noether current of $\mathcal{L}_{-m,-\lambda}$ is $i\sqrt{|g|} j^\mu$ (M1, item 2, with $(-m, -\lambda)$), so the charge of the second field is $Q_- = -Q_+$ at every x_4 and $Q_+ + Q_- = 0$. By M1 each charge is conserved, $dQ_\pm/dx_4 = 0$. By H3, $B_\pm = Q_\pm$, and by H2, $B_+ \neq 0$. The statement about $T_{\mu\nu}^{\text{pair}}$ is Corollary M4.1. $\square$

The Wolfram verifier checks the implication symbolically: $Q_+ + Q_-$ simplifies to 0, and dQ_-/dx_4 simplifies to 0 given $dQ_+/dx_4 = 0$ (check MA_M5_implication, Wolfram only, with the ingredients computed in the checks of M1 and M4). The Python checker records the implication and its ingredients as the measurement M5_conditionalScenario, without a separate check. The implication is elementary; the substance lies in the hypotheses.

8.3 What M5 does not do

1. It does not produce an asymmetry. By M1, Q_+ is constant, so a nonzero Q_+ today is the same as a nonzero Q_+ at $x_4 = 0$: the asymmetry of our universe is an initial condition. H2 puts it in by hand.
2. It does not predict η . The observed $\eta \approx 6 \times 10^{-10}$ is **not predicted**: nothing here computes the magnitude or the sign of Q_+ , nor the photon content that η refers to.
3. It does not satisfy Sakharov's conditions; it replaces them by an assumed correlated configuration of two classical fields, a global symmetry of the pair, together with an initial condition in each member.
4. It is classical. The balance holds for two classical fields in the correlated configuration of H1. At the quantum level no reading gives a cancellation between two independent, consistently quantised universes (Section 7.3). The image field $\gamma^8 \Psi_+$ is the same quantum system as Ψ_+ , with its own energy, charge and gravitational source $+H_+$, $+Q_+$ and $+T_+$. An independently quantised $-M$ universe has positive kinetic and mass energies that add to those of the $+M$ universe, and its charge would cancel only by a further assumption about its state.
5. It is not a compensation by a second universe. Even for the classical fields, $Q_- = -Q_+$ is a property of the assumed configuration, not the result of a process that moves charge from one member to the other. In the image-field reading of the quantum theory the $-M$ member has no degrees of freedom of its own, $Q_+ + Q[\gamma^8 \Psi_+] = 0$ is an identity valid in every state, including every state with $Q_+ \neq 0$, and "created together" has no content.
6. It does not relate the dirac16complex charge to the baryons of the Standard Model (H3), and it does not show that pairs are created (H1).

9 M6: the Sakharov scorecard

Sakharov condition [1]	Status in the theory as built	What would have to be added
1. Violation of the conserved charge (applied to the U(1) charge Q , the only candidate; Q stands for baryon number only under H3)	Fails, by M1: Q is exactly conserved in every gravitational field (vanishing boundary flux), for both statistics and every potential	A U(1)-violating interaction from the list of M3 (Section 10, item 1)
2. C and CP violation	Fails for the Lagrangian, by M2: C is exact for the commuting field in every gravitational field, and CP is exact for the anticommuting field in flat space and in gravitational fields with the reflection isometry; each reverses Q and preserves x_4 . A generic gravitational field breaks CP of the anticommuting field, which does not help because row 1 fails	Terms that break all 64 exact charge-reversing symmetries without x_4 reversal (Section 10, item 2)
3. Departure from thermal equilibrium	Not addressed by any computation; the Kohn-Sham states of Stages 4 and 5 are equilibrium states at fixed N	A non-equilibrium history and a computation of rates (Section 10, item 3)

Since condition 1 fails exactly, no departure from equilibrium can create a net charge inside one universe. The computed checks behind the rows are

```

row 1  Wolfram  MA_M1_noetherIdentity_grassmann_G1  MA_M1_noetherIdentity_commuting_G1
        Python  MA_M1_noetherIdentity_<X>_<G> for both statistics and both geometries
row 2  Wolfram  MA_M2_symmetrySummaryAndChargeReversal
        Python  MA_M2_discreteGroupCharacterTable  MA_M2_C_and_CP_status
                MA_M2_cpScopeInCurvedFields
row 3  none; nothing was computed

```

10 What would have to be added to the theory

The following list is a consequence of M1 to M3. It describes necessary ingredients; none of them is present in the theory, none is claimed to be natural, and none has been shown to be sufficient.

1. **A charge-violating interaction.** By M3, for the anticommuting field no Majorana-type mass term exists. The lowest-order candidates are the derivative term $\sqrt{|g|} \Psi^T C \gamma^8 \gamma^\mu D_\mu \Psi$ plus its Hermitian conjugate (charge 2), and quartic terms such as Q_4 (charge 4) and Q_2 (charge 2) plus their conjugates. For the commuting field the mass terms $\Psi^T C P_\pm \Psi$ and

the derivative term $\sqrt{|g|} \Psi^T C \gamma^\mu D_\mu \Psi$ (the form of the notebook Lagrangian Lg[]), each plus its complex conjugate, have charge 2. Each such term breaks the U(1) of M1 to a discrete subgroup (derived): $e^{i\alpha}$ with $e^{2i\alpha} = 1$ for charge-2 terms, $e^{4i\alpha} = 1$ for charge-4 terms.

2. **Breaking of every charge-reversing symmetry.** The added terms must break every exact symmetry of Section 5 that reverses Q without reversing x_4 : C_0 (and the 63 others) for the commuting field, the CP family for the anticommuting field (64 maps, among them $C_8 P_b$ for $b = 0, 1, 2, 3$ and $C_8 P_{123}$). A remark that is derived but not machine-checked: the U(1) rotation $\Psi \rightarrow e^{i\alpha} \Psi$ is a symmetry of $\mathcal{L}$ and multiplies the coefficient g of a charge-2 term by $e^{2i\alpha}$, so the phase of a single such coefficient can be removed; only relative phases between several charge-violating terms can be physical. Whether a given combination breaks the CP family is not computed here.
3. **A departure from equilibrium.** A dynamical history in which the charge-violating processes are out of equilibrium, for example during the evolution of the primordial field, and a computation of their rates. Nothing of this kind has been computed.
4. **A link to baryons.** The dirac16complex charge would have to be connected to the baryon number of the Standard Model (H3), which requires couplings to Standard-Model fields that the theory does not contain.
5. **A computation of η .** Only with items 1 to 4 could a value of η be computed and compared with $\eta \approx 6 \times 10^{-10}$ [2], [5].

11 Verification records

11.1 The reports

Two independent exact implementations were written for this analysis, and two Stage-5 reports are used as inputs. All four have schemaVersion 1, and every check in them is true:

```
artifacts/dirac16complex/matter-antimatter/wolfram-matter-antimatter-report.json
44 of 44 checks true
producer scripts/verify_dirac16complex_matter_antimatter.wls
    with wolfram/Dirac16ComplexMatterAntimatter.wl, Wolfram Language 15.0.1
also writes artifacts/dirac16complex/matter-antimatter/matter-antimatter-theory.json
artifacts/dirac16complex/matter-antimatter/python-matter-antimatter-report.json
77 of 77 checks true
producer scripts/check_dirac16complex_matter_antimatter.py
    with scripts/grassmann_algebra.py (the report records no versions;
    the runs of Section 11.3 used Python 3.14.5, sympy 1.14.0, numpy 2.4.6)
artifacts/dirac16complex/pair-creation/wolfram-pairing-report.json (Stage-5 input)
141 of 141 checks true
producer scripts/verify_dirac16complex_pairing.wls (wolfram/Dirac16ComplexPairing.wl)
artifacts/dirac16complex/pair-creation/python-pairing-report.json (Stage-5 input)
172 of 172 checks true
producer scripts/check_dirac16complex_pairing.py
```

Both matter-antimatter implementations compute everything exactly: integers, rationals, Gaussian rationals, exact symbolic algebra and exact Grassmann algebras. No floating-point number decides

any check, with the one labelled exception MA_M1_ksFixedNetNumberRecorded (Section 4.4). Each checker prints `check_<name>=true|false`, `measurement_<name>=...`, `check_count` and `failed_check_count`, and exits with a nonzero status if a check fails. Both start from the gamma matrices built from the definitions and compare them with the exact fixture: the Wolfram package compares η , $\gamma^0, \dots, \gamma^7$, C , γ^8 , the 28 S^{ab} , B and the gammas of the Stage-1 geometry package (MA_algebra_fixtureMatches, MA_algebra_basicFacts); the Python checker compares the gammas, C , the chirality, η , the S^{ab} and B (MA_fixtureAgreement). The Python checker uses nothing produced by Wolfram as truth; it compares with the Wolfram theory file only in its agreement checks (Section 11.3).

11.2 All checks

The Wolfram report, all 44 checks:

```
MA_algebra_fixtureMatches MA_algebra_basicFacts
MA_M1_u1InvarianceCommutingGenericU_flat MA_M1_u1InvarianceCommutingGenericU_G1
MA_M1_u1InvarianceGrassmann_G1 MA_M1_grassmannPotentialsPolynomialAndNeutral
MA_M1_noetherCurrentLocalPhase_commuting_G1
MA_M1_noetherCurrentLocalPhase_grassmann_G1 MA_M1_noetherCurrentFormula_grassmann_G1
MA_M1_noetherCurrentFormula_commuting_G1 MA_M1_noetherIdentity_grassmann_G1
MA_M1_noetherIdentity_commuting_G1 MA_M1_negativeControlNotebookConnection
MA_M1_onShellConservation_G1 MA_M1_chargeDensityMatrix MA_M1_ksFixedNetNumberRecorded
MA_M1_stage1ChecksCited MA_M2_conjugationIntertwiners MA_M2_transposeIntertwiners
MA_M2_signPatternClassification MA_M2_statisticsSign MA_M2_matrixClassification
MA_M2_lagrangianFlatCommuting_all MA_M2_lagrangianFlatGrassmann_all
MA_M2_namedTransformations MA_M2_frameLevelG1_commuting MA_M2_frameLevelG1_grassmann
MA_M2_symmetrySummaryAndChargeReversal MA_M2_canonicalStructure
MA_M3_spinInvariantForms MA_M3_pinCharacterForms MA_M3_kineticInvariantForms
MA_M3_grassmannSurvival MA_M3_commutingSurvival MA_M3_u1Charge
MA_M3_pinCharactersMajoranaTerms MA_M3_notebookLgIsMajoranaType
MA_M3_extraGrassmannQuarticCharge4 MA_M4_currentFlipMatrix
MA_M4_currentFlipG1_grassmann MA_M4_pairEMTAndCurrentG1_commuting
MA_M4_pairEMTG1_grassmann MA_M4_kreinOneParticle MA_M5_implication
```

The Python report, all 77 checks:

```
MA_fixtureAgreement MA_M1_geometry_G_A_generic_nondiagonal
MA_M1_u1Invariance_grassmann_G_A_generic_nondiagonal
MA_M1_noetherCurrentIsISqrtgPsibarGammaPsi_grassmann_G_A_generic_nondiagonal
MA_M1_eulerLagrangeCrossCheck_grassmann_G_A_generic_nondiagonal
MA_M1_eulerLagrangeClosedForm_grassmann_G_A_generic_nondiagonal
MA_M1_noetherIdentity_grassmann_G_A_generic_nondiagonal
MA_M1_currentDivergenceIdentity_grassmann_G_A_generic_nondiagonal
MA_M1_hermiticity_grassmann_G_A_generic_nondiagonal
MA_M1_u1Invariance_commuting_G_A_generic_nondiagonal
MA_M1_noetherCurrentIsISqrtgPsibarGammaPsi_commuting_G_A_generic_nondiagonal
MA_M1_eulerLagrangeCrossCheck_commuting_G_A_generic_nondiagonal
MA_M1_eulerLagrangeClosedForm_commuting_G_A_generic_nondiagonal
MA_M1_noetherIdentity_commuting_G_A_generic_nondiagonal
MA_M1_currentDivergenceIdentity_commuting_G_A_generic_nondiagonal
```

```

MA_M1_hermiticity_commuting_G_A_generic_nondiagonal MA_M1_geometry_G_B_diagonal_x0_x4
MA_M1_u1Invariance_grassmann_G_B_diagonal_x0_x4
MA_M1_noetherCurrentIsISqrtgPsibarGammaPsi_grassmann_G_B_diagonal_x0_x4
MA_M1_eulerLagrangeCrossCheck_grassmann_G_B_diagonal_x0_x4
MA_M1_eulerLagrangeClosedForm_grassmann_G_B_diagonal_x0_x4
MA_M1_noetherIdentity_grassmann_G_B_diagonal_x0_x4
MA_M1_currentDivergenceIdentity_grassmann_G_B_diagonal_x0_x4
MA_M1_hermiticity_grassmann_G_B_diagonal_x0_x4
MA_M1_u1Invariance_commuting_G_B_diagonal_x0_x4
MA_M1_noetherCurrentIsISqrtgPsibarGammaPsi_commuting_G_B_diagonal_x0_x4
MA_M1_eulerLagrangeCrossCheck_commuting_G_B_diagonal_x0_x4
MA_M1_eulerLagrangeClosedForm_commuting_G_B_diagonal_x0_x4
MA_M1_noetherIdentity_commuting_G_B_diagonal_x0_x4
MA_M1_currentDivergenceIdentity_commuting_G_B_diagonal_x0_x4
MA_M1_hermiticity_commuting_G_B_diagonal_x0_x4 MA_M1_divergenceIdentityAllFirstJets
MA_M1 MA_M2_chargeConjugationSolutionSpaces
MA_M2_psibarTransposeFormsReduceToPsiStarForms MA_M2_internalMapsCurvedJets
MA_M2_internalMapsAsDerived MA_M2_cpScopeInCurvedFields
MA_M2_namedReflectionsFlatJets
MA_M2_genericUnitVectorReflections MA_M2_reflectionsInCurvedFieldWithIsometry
MA_M2_spin0ContainsChargeReversingTimeRotation MA_M2_discreteGroupCharacterTable
MA_M2_characterHomomorphism MA_M2_C_and_CP_status
MA_M2_canonicalStructureOfExactGrassmannSymmetries
MA_M2_chargeConjugationWithSpatialParity MA_M2 MA_M3_invariantFormsSpan_C_Cgamma8
MA_M3_allInvariantFormsSymmetric MA_M3_fullSpinCharacter MA_M3_pinCharacters
MA_M3_derivativeBilinearClassification MA_M3_massTypeSurvivalAndCharge
MA_M3_derivativeTypeSurvivalCurved MA_M3_quarticChargeViolatingExamplesGrassmann
MA_M3 MA_M4_gamma8MatrixFacts MA_M4_gamma8ChargeFlipCurved
MA_M4_gamma8EulerLagrangePairing MA_M4_gamma8EMTPairing MA_M4_kreinModeFacts
MA_M4_imageFieldFockModel MA_M4 MA_wolfram_chargeConjugationIntertwiners
MA_wolfram_classificationRows
MA_wolfram_symmetrySummary MA_wolfram_chargeReversingSymmetries
MA_wolfram_invariantForms MA_wolfram_pinCharacters MA_wolfram_kineticForms
MA_wolfram_survival MA_wolfram_quarticQ4 MA_wolfram_kreinOneParticle
MA_wolfram_gamma8MatrixFacts MA_agreesWithWolfram MA_internal_noException

```

The honest answer is recorded in the reports themselves: in the Python report as the measurement `honestAnswer`, and in the Wolfram theory file as the key `honestyRule`. Both state that the claim is not proved and not provable within the theory as built.

11.3 Agreement between the implementations

The Python check `MA_agreesWithWolfram` compares its own results with the Wolfram theory file item by item: the charge-conjugation intertwiners (proportional bases), the classification rows (2048 compared, no disagreement), the C, P, CP, T and CPT summary (22 items compared, no difference), the number of exact charge-reversing symmetries without x_4 reversal (64 for each statistics in both), the invariant forms, the Pin-covariant forms, the kinetic forms, the survival table, the quartic Q_4 (40 monomials in both, proportional with factor 8), the one-particle Krein facts (including the canonical generators of the image field) and the γ^8 matrix facts. The two implementations were written independently and share only the fixture and the conventions.

The Python report records in its inputSha256 the hash of the Wolfram theory file it compared with, and that hash is the hash of the current theory file:

```
matter-antimatter-theory.json compared by the Python report
1505a3962938477e5db42f568c0a1afcd21898c4409793178f8311b792638f10
```

After the last change of either implementation both were run again. The Wolfram verifier was run twice, once into the committed paths and once with the report and theory paths in a scratch directory; each run took about 12 minutes and reported 44 of 44 checks true, and the two runs wrote byte-identical reports and byte-identical theory files, equal to the committed ones. The Python checker was run twice against that theory file (about 70 seconds each); both runs reported 77 of 77 checks true and wrote byte-identical reports, equal to the committed one.

11.4 Stage-5 inputs

What is proved here and what is cited. The field-level and one-particle statements of M4 (Sections 7.1 and 7.2) are proved here, by both implementations. The canonical structure of the image field and the operator identities of Section 7.3 in the four-mode Fock model are proved here by the Python implementation only (MA_M4_imageFieldFockModel). The remaining Fock-level statements of Section 7.3 (the Stage-5 Fock model with the sea and normal ordering, the normal-ordered energy-momentum tensor, the Stage-5 mode list) are cited from the Stage-5 exact pairing results: artifacts/dirac16complex/pair-creation/pairing-theory.json (key T1krein), the Stage-5 Wolfram report wolfram-pairing-report.json (141 of 141 checks true, among them the twelve PAIR_T1krein checks) and the independent Stage-5 Python report python-pairing-report.json (172 of 172 checks true, among them the fourteen S5_T1krein checks, compared against the same pairing-theory.json). The Stage-5 reports are complete, but the Stage-5 documents and the Stage-5 gate are not, and the files carry no finality flag. Both matter-antimatter reports therefore record the citation as PROVISIONAL, open until the Stage-5 gate passes (Wolfram measurement M4_kreinLevelStatus, Python measurement stage5Pairing). The hashes of pairing-theory.json and of the Stage-5 Wolfram report are recorded in both matter-antimatter reports, the hash of the Stage-5 Python report in the Python report. The Stage-5 text T1krein.imageField describes the image as a universe with energy $-|\epsilon|$ and charge -1 per quantum; Section 7.3 explains why these are the values of the $\mathcal{L}_{-m,-\lambda}$ formulas and not the image field's own energy and charge. If the Stage-5 files change, both matter-antimatter verifiers have to be rerun.

11.5 Files and hashes

The sha256 values of the producers and inputs as recorded in the reports, and of the two Wolfram outputs (the Stage-5 Python report is an input of the Python checker only):

```
wolfram/Dirac16ComplexMatterAntimatter.wl
382c34694f91bf2ed68cee42736dd9298171e2abf61d0b388d6228e4d47ef4c5
scripts/verify_dirac16complex_matter_antimatter.wls
5d85f7d8a76a78b0f8853d41af0d2a352b426c553d5eba4569ade2fd5d9fc73f
wolfram/Dirac16ComplexGeometry.wl
f5b674665eee4000750161e6ab6312c38bfac9da7a17450a2b3bdd88ef292af2
```

```

scripts/check_dirac16complex_matter_antimatter.py
  ef1a00dc02e72186cd6c78d83ad3d930c372ccd9440983a307e8e04cae622231
scripts/grassmann_algebra.py
  dfce851d3dc7ab82202a62d414cc259644a64a101f08e9ba80ed747faa8bc24c
artifacts/dirac16complex/arbitrary-field/algebra-fixture.json
  8b4f15462ca4d61ef6ec0e72f04c8a77d23ce8bcabc9b6ce19f921c90e02653b
artifacts/dirac16complex/pair-creation/pairing-theory.json
  5a267bd696391b131134577ccdcf7766b83f96f3eb32b9b8de60c9175b1ebf6a
artifacts/dirac16complex/pair-creation/wolfram-pairing-report.json
  735534de950c7fb0327370c33caa275cf805aa91a41dee903e4eaec7a6fa0de8
artifacts/dirac16complex/pair-creation/python-pairing-report.json
  773b3a63aa041ad873c31635cb69b36f548c40db47fcff68d6603da5e38fcc97
artifacts/dirac16complex/matter-antimatter/matter-antimatter-theory.json
  1505a3962938477e5db42f568c0a1afcd21898c4409793178f8311b792638f10
artifacts/dirac16complex/matter-antimatter/wolfram-matter-antimatter-report.json
  ad6b91296034a24c14b2ee08c7bfe8da98aaff2db83544897767e498adeaa269

```

The Stage-1 reports cited in MA_M1_stage1ChecksCited:

```

artifacts/dirac16complex/arbitrary-field/wolfram-geometry-report.json
  cec9ee0d577c4e7f0e82efd404d412a503140ccb1547118ab48a78a3d773f0ac
artifacts/dirac16complex/arbitrary-field/wolfram-algebra-report.json
  d43adeeba580bdd8cec56fac5fa74906b7578c06c8589e52dbe3da1404cd49e7
artifacts/dirac16complex/arbitrary-field/grassmann-demo-report.json
  b83997857ccfc9fb71f05efbcaebe16f29bf36687cc08c16e0823a26a6369f09

```

12 Reproduction

Run every command from the repository root. The order matters: the Wolfram verifier reads the Stage-5 pairing files and writes both its report and matter-antimatter-theory.json, and the Python checker compares with that theory file. The Wolfram report path is a plain positional argument, because WolframScript 1.14 drops a double-hyphen separator and every argument after it when it is combined with -file; an optional second positional argument sets the theory path. The Python checker takes its report path from its output option, as in the scratch rerun at the end of Section 12.2. Set PYTHONUTF8=1 for the Python steps. The Wolfram step takes about 12 minutes, the Python checker about a minute.

12.1 PowerShell

```

$env:PYTHONUTF8 = "1"
$m = "artifacts/dirac16complex/matter-antimatter"
$v = "scripts/verify_dirac16complex_matter_antimatter.wls"
wolframscript -file $v "$m/wolfram-matter-antimatter-report.json"
python scripts/check_dirac16complex_matter_antimatter.py
python -m unittest discover -s tests -p "test_d16c_matter_antimatter.py" -v
python scripts/build_provenance_pdf.py provenance/DIRAC16COMPLEX_MATTER_ANTIMATTER.md
python -m unittest discover -s tests -p "test_d16c_matter_antimatter_publication.py" -v

```

12.2 Git Bash

```
export PYTHONUTF8=1
m=artifacts/dirac16complex/matter-antimatter
wolframscript -file scripts/verify_dirac16complex_matter_antimatter.wls \
    "$m/wolfram-matter-antimatter-report.json"
python scripts/check_dirac16complex_matter_antimatter.py
python -m unittest discover -s tests -p "test_d16c_matter_antimatter.py" -v
python scripts/build_provenance_pdf.py provenance/DIRAC16COMPLEX_MATTER_ANTIMATTER.md
python -m unittest discover -s tests -p "test_d16c_matter_antimatter_publication.py" -v
```

A rerun that leaves the committed files untouched writes into a scratch directory (here build/, which git ignores):

```
wolframscript -file scripts/verify_dirac16complex_matter_antimatter.wls \
    build/ma/wolfram-report.json build/ma/theory.json
python scripts/check_dirac16complex_matter_antimatter.py \
    --theory build/ma/theory.json --output build/ma/python-report.json
```

The Stage-5 inputs are regenerated, if needed, before the first step with

```
wolframscript -file scripts/verify_dirac16complex_pairing.wls \
    artifacts/dirac16complex/pair-creation/wolfram-pairing-report.json
```

(in PowerShell each of these commands is written on one line, without the backslash). If this changes the bytes of the pairing files, their hashes in Section 11.5 change as well. The PDF step builds this document twice (two builder runs, three pdflatex passes each), requires warning-free logs and byte-identical PDFs, and compares the result with the registered edition dirac16complex-matter-antimatter in provenance/pdf-specifications.json. After an edit of this document the edition is registered again with

```
python scripts/build_provenance_pdf.py --register \
    provenance/DIRAC16COMPLEX_MATTER_ANTIMATTER.md
```

(one line in PowerShell), and the sha256 pins in tests/test_d16c_matter_antimatter_publication.py are updated. That test also checks that the check counts, check names, numbers and hashes quoted here agree with the reports.

13 Non-claims

1. This document does not claim, and does not prove, that the dirac16complex theory solves the matter-antimatter problem or any of the current matter-antimatter mysteries. Within the theory as built the claim is false in its central part (M1, M2).
2. The observed baryon-to-photon ratio $\eta \approx 6 \times 10^{-10}$ is not predicted, and no value of the charge of our universe is computed.

3. No creation of universes is derived: no creation process, rate, amplitude or wave function of the universe. Corollary M4.1 is a statement about conservation laws and constraints.
4. H1, H2 and H3 are hypotheses. The classical-level implication of M5 is proved; the scenario is not a result, and it has no quantum counterpart with two independent universes (Section 7.3).
5. The `dirac16complex` charge is not baryon number. The theory contains no quarks, leptons or baryons of the Standard Model.
6. The $-M$ member of the pair is not antimatter in our universe. Its negative mass is a parameter of a second field configuration; nothing here says that antimatter has negative mass or falls upward, which the ALPHA-g observation [12] rules out for antihydrogen.
7. The charge-violating terms of M3 are a classification of what the symmetry allows. They are not proposed as part of the theory, they are not claimed to be natural, and higher-order terms are not classified.
8. The quantum statements are formal: the canonical state space is a Krein space, the positive Fock space exists only in the good sector, and the extra-time sector is ill-posed (Stage 1, Section 10.10). The conservation of the quantized charge is argued without a regularization, and no anomaly is computed (Section 4.2). The discrete maps of Section 5.6 are automorphisms of the anticommutation relations; their implementation on a Fock space is not decided. The commuting field `dirac16complex00` is treated as a classical field.
9. The signature (4,4) is not the signature of observed spacetime, and nothing here is fitted to or compared with data.
10. Nothing here tests or depends on the CPT-symmetric universe of Boyle, Finn and Turok [3]; it is cited as the known published example of a pair-of-universes idea.
11. The check `MA_M1_ksFixedNetNumberRecorded` is a record of floating-point data read from the committed Stage-4 runs, not a proof.

14 References

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